

ContentForge: An AI-Driven Content Creation Platform for Digital Media Using Gemini-Powered Generation

Project Report

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[GITS LOGO]

Darkness to Light

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CERTIFICATE

This is to certify that the work embodied in this Project entitled "ContentForge: An AI-Driven Content Creation Platform for Digital Media Using Gemini-Powered Generation and LangChain Orchestration" has been carried out by Chhavi Vinaik under my guidance and supervision for partial fulfilment of the requirement of the award of the Degree of Bachelor of Technology in Computer Science and Engineering of Rajasthan Technical University, Kota (Raj.) during the academic year 2024-25. The matter presented in this project has not been submitted for the award of any other degree of this or any other university.

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I, Chhavi Vinaik, student of M.Tech (CSE), Geetanjali Institute of Technical Studies, Dabok, Udaipur (Raj.) do hereby declare that the Project entitled "ContentForge: An AI-Driven Content Creation Platform for Digital Media Using Gemini-Powered Generation and LangChain Orchestration" is my own work conducted under the supervision of Dr. Ajay Kumar Sharma, Associate Professor, Department of CSE, GITS, Dabok.

I further declare that to the best of my knowledge, the Project does not contain any work which has been submitted for the award of any degree or diploma in any university or institution. Any information derived from other sources has been duly acknowledged in the project.

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ABSTRACT

The rapid expansion of digital media platforms has created an increasing demand for scalable, personalised, and high-quality content across text and multimedia formats. Traditional manual content creation workflows are unable to meet the speed, volume, and adaptability required by modern ecosystems such as social media platforms, digital marketing channels, online journalism, and creator-driven environments. Recent advances in Artificial Intelligence, particularly Large Language Models (LLMs), have enabled automated content generation through natural language prompts, transforming AI into an active participant in the content creation lifecycle. Models with few-shot and zero-shot learning capabilities can generate articles, captions, scripts, and promotional material without task-specific retraining, making them suitable for real-world deployment.

This project proposes ContentForge — an application-oriented AI-driven content creation framework designed specifically for digital media platforms, emphasising simplicity, modularity, and deployability over architectural complexity. The system allows users to provide prompts through a centralised interface, after which content is generated using Google Gemini 2.5 Flash as the core generative model and refined through LangChain-managed workflows. The architecture incorporates automated quality evaluation, lightweight personalisation, iterative refinement, responsible AI safety filtering, and AI-powered image prompt synthesis to ensure outputs are relevant, safe, and platform-ready. In parallel, orchestration frameworks such as LangChain enable structured prompt handling, iterative refinement, and contextual memory management.

The backend is built with FastAPI using an asynchronous background-task model, SQLAlchemy ORM for persistence, and Docker for containerisation. The frontend is built with Next.js 14 and TypeScript, implementing a polling-based real-time update mechanism. Experimental observations indicate that the proposed approach significantly reduces content production time while maintaining coherence, relevance, and stylistic consistency. By integrating controlled generation, feedback-driven refinement, and responsible AI safeguards, this framework demonstrates the feasibility and viability of AI-driven content creation as a scalable solution for next-generation digital media platforms.

Keywords: AI-Driven Content Creation, Digital Media Platforms, Large Language Models, Generative AI, LangChain, Google Gemini, Prompt Engineering, Content Personalisation, Responsible AI, FastAPI, Next.js, SQLAlchemy.

Chapter-1

INTRODUCTION

1.1 Background and Motivation

The modern digital media landscape operates at a pace and scale that was unimaginable a decade ago. Thousands of articles, social media posts, marketing campaigns, and video scripts are published every hour across platforms that span social networks, news portals, e-commerce channels, and creator communities. This relentless demand for fresh, high-quality, and contextually relevant content has created a structural challenge for organisations and individuals who rely on human writers and creative teams.

Simultaneously, breakthroughs in Natural Language Processing — particularly the emergence of transformer-based Large Language Models — have opened a new frontier in automated content creation. These models can generate coherent, stylistically diverse, and contextually appropriate text from simple natural language prompts. However, the gap between research-grade LLM capabilities and production-ready deployment remains wide. ContentForge bridges this gap by providing a structured, modular, and deployable AI-driven content creation platform for digital media use cases.

1.2 Growth of Digital Media and Content Demand

Digital media platforms have evolved into fast-paced ecosystems where content is produced and consumed continuously. Social networks, digital news portals, marketing channels, and creator-based applications demand fresh, engaging, and personalised content at scale. Traditional content creation methods, which rely heavily on manual writing and design, struggle to match this speed and volume. As audience expectations rise, platforms require automated solutions that can generate consistent, high-quality content while adapting to trends, formats, and user preferences in real time.

Industry surveys indicate that content teams spend over 60% of their time on repetitive drafting and editing tasks that could be automated. The global content marketing industry is

valued at over USD 600 billion, with an acute shortage of skilled content creators who can produce platform-specific, audience-targeted content at scale. This demand gap makes AI-driven content creation not merely a convenience but a strategic necessity for digital media operations.

1.3 Role of Large Language Models in Content Creation

Large Language Models such as GPT-4 and Google Gemini have demonstrated remarkable ability in generating text with coherence, contextual relevance, and stylistic versatility. These models leverage few-shot and zero-shot learning to produce content without task-specific fine-tuning, making them suitable for diverse digital media applications. For content creation, LLMs provide a scalable mechanism for automating articles, captions, marketing copy, scripts, and summaries across multiple languages and tones.

Unlike earlier rule-based or template-driven systems, LLMs understand the semantic intent behind a prompt, adapt tone and register to the specified platform, and generate outputs that are indistinguishable in quality from human-written content when properly guided. The challenge lies in operationalising these capabilities — ensuring safety, quality, personalisation, and platform alignment — within a reliable, auditable pipeline.

1.4 LangChain and Orchestration Frameworks

Orchestration frameworks such as LangChain enable structured prompt handling, iterative refinement, and contextual memory management, allowing AI systems to move beyond single-step generation toward interactive and adaptive content creation pipelines. LangChain provides chain composition primitives (LCEL — LangChain Expression Language), prompt template management, structured output parsing, and tool integration capabilities that make it possible to build deterministic multi-stage workflows.

In ContentForge, LangChain chains connect individual Gemini calls into a coherent pipeline where each stage builds on the output of the previous one. Prompt templates are version-controlled in `templates.py` and parameterised with user inputs, ensuring consistent and testable behaviour across the generation pipeline. The `StrOutputParser` and custom JSON extraction utilities ensure reliable structured outputs from probabilistic LLM responses.

1.5 Shift from Research to Application-Based Systems

While generative AI research has produced highly complex architectures, real-world digital media platforms require practical, application-oriented systems. Application-based AI content creation focuses on usability, refinement, and deployment rather than theoretical complexity. Users expect intuitive prompt-based interfaces that generate ready-to-use outputs with minimal effort. This shift emphasises modular pipelines, automated refinement, and platform compatibility.

ContentForge adopts this application-first philosophy. Rather than exposing the underlying model complexity, the system abstracts it behind a simple natural language interface. Users specify what they want, and the system handles prompt engineering, quality assurance, personalisation, and image generation internally — making AI content creation accessible to non-technical creators, marketers, and journalists.

1.6 Need for Responsible and Scalable AI Content Systems

As AI-generated content becomes widespread, concerns related to content quality, bias, misinformation, and ethical use must be addressed. Responsible AI practices such as safety filtering, bias mitigation, and controlled generation are essential for maintaining trust in digital platforms. At the same time, systems must be scalable to support growing user bases and diverse content needs.

ContentForge integrates responsibility at the pipeline entry point through a two-layer safety filter (keyword blocklist + Gemini classifier) and at the output level through quality scoring and human-readable feedback. The asynchronous FastAPI architecture handles concurrent requests without blocking, ensuring scalability as the user base grows. Docker containerisation and Azure Pipelines CI/CD make the system deployable and maintainable at production scale.

1.7 Project Overview: ContentForge

ContentForge is an application-oriented AI-driven content creation framework that integrates Google Gemini 2.5 Flash as the core language model with LangChain as the orchestration backbone. The system processes user prompts through an 8-stage pipeline: (1) Safety Filtering, (2) Prompt Refinement, (3) Initial Content Generation, (4) Quality Evaluation, (5) Iterative Rewriting, (6) Personalisation, (7) Image Prompt Synthesis, and (8) Image Rendering.

The backend is implemented in Python using FastAPI with asynchronous background task processing and SQLAlchemy ORM for generation history persistence. The frontend is a Next.js 14 application with TypeScript and Tailwind CSS, implementing a polling-based real-time status update mechanism. The platform targets content creators, marketers, journalists, and educators who need high-quality, platform-ready content at scale without requiring AI expertise.

1.8 Technology Stack Summary

Layer	Technology	Purpose
Frontend	Next.js 14, TypeScript, Tailwind CSS	Responsive UI, real-time polling, content display
Backend	FastAPI, Python 3.10, Uvicorn	Async REST API, background tasks, CORS, static files
AI Engine	Google Gemini 2.5 Flash	Content generation, safety classification, quality evaluation, rewriting, personalisation
Orchestration	LangChain, langchain_google_genai	Pipeline chaining (LCEL), prompt templates, StrOutputParser
Database	SQLite via SQLAlchemy ORM	Content history, generation records, refinement logs

Image Gen	Pollinations API (Flux) / Gemini Imagen	AI-powered editorial image generation
Configuration	pydantic-settings, .env files	Type-safe environment variable management
DevOps	Docker, docker-compose, Azure Pipelines	Containerisation, CI/CD, reproducible deployment

Table 1.1 — Technology Stack of ContentForge

1.9 Report Organisation

This report is organised across 13 chapters. Chapter 2 presents the problem statement; Chapter 3 defines objectives and scope; Chapter 4 provides a detailed Software Requirement Specification; Chapter 5 describes the project development lifecycle; Chapter 6 covers system planning, risk assessment, and the PERT chart; Chapter 7 presents all 12 system design diagrams as Mermaid code blocks; Chapter 8 covers technical implementation with actual code snippets from the codebase; Chapter 9 describes the testing approach and results; Chapter 10 presents project screenshots; Chapter 11 concludes with future work; Chapters 12 and 13 list references and publication details respectively.

1.9.1 How to Read This Report

This report is structured to support both sequential reading and topic-based reference. Readers with a software engineering background who are primarily interested in the technical architecture may focus on Chapters 7, 8, and 9, which cover system design, implementation, and testing in detail. Readers interested in the research context and literature positioning should read Chapters 1 and 2, which provide the theoretical background and problem statement. Project managers and academic evaluators assessing the project methodology and planning should consult Chapters 5 and 6, which document the development lifecycle and system planning. Chapters 3 and 4

(Objectives, SRS) provide the formal requirements specification and are particularly relevant for evaluators assessing requirement completeness and traceability.

Code listings in Chapter 8 are presented as pseudocode-style Python excerpts that prioritise readability over syntactic completeness. Readers who wish to examine the complete, executable source code should refer to the project repository. UML and architecture diagrams in Chapter 7 are presented as Mermaid diagram code blocks; the rendered diagram descriptions accompanying each code block provide a textual interpretation suitable for readers who are not familiar with UML notation.

1.10 Literature Review and Related Work

A critical examination of the existing body of literature is essential to understand the state of the art in AI-driven content generation, identify gaps, and justify the design choices made in ContentForge. This section surveys commercial platforms, academic research, orchestration frameworks, quality evaluation techniques, and responsible AI practices, situating ContentForge within this broader landscape.

1.10.1 Commercial AI Content Generation Platforms

Several commercial platforms have emerged to address the growing demand for automated content creation. Jasper (formerly Jarvis), Copy.ai, Writesonic, and Rytr are among the most widely adopted, each offering template-driven content generation powered by large language models. These platforms rely primarily on single-pass generation: a user enters a prompt, the platform calls an LLM API, and the result is returned without further quality validation or iterative refinement. While these tools have demonstrated commercial viability, academic and industry evaluations consistently reveal their limitations: inconsistent quality across content types, lack of platform-specific adaptation, absence of transparent quality scoring, and no built-in responsible AI filters beyond those provided by the underlying model. ContentForge differentiates itself from these platforms by embedding a multi-stage quality assurance loop and

a dedicated two-layer safety system directly within its generation pipeline, rather than relying on model-level guardrails alone.

Additionally, platforms such as HubSpot AI, Canva Magic Write, and Adobe Firefly have integrated generative AI into broader creative suites, combining text and image generation. However, these integrations remain proprietary, closed-source, and tightly coupled to their respective ecosystems. ContentForge, by contrast, provides an open, extensible, API-first architecture that can serve as a reusable backend for any frontend or CMS integration.

1.10.2 Large Language Models for Content Generation: Research Perspective

The foundational capability enabling AI content generation is the transformer-based Large Language Model. Brown et al. (2020) demonstrated that GPT-3, with 175 billion parameters, could perform few-shot content generation tasks with human-comparable quality, establishing the viability of LLMs as content production engines. Subsequent work by Cao et al. (2023) provided a comprehensive survey of AI-Generated Content (AIGC) across modalities, identifying prompt engineering, chain-of-thought reasoning, and fine-tuning as the primary techniques for improving generation quality. The emergence of instruction-tuned models such as GPT-4 and Google Gemini 2.5 further elevated the baseline quality of generated text, reducing the need for extensive prompt engineering while enabling richer contextual adaptation.

Zhang et al. (2024) studied guided profile generation for LLM-based personalisation, demonstrating that structured intermediate representations (analogous to ContentForge's ContentBrief dataclass) significantly improve the relevance and tone-consistency of generated outputs. Their findings directly informed the design of ContentForge's prompt refiner stage, which converts raw user prompts into structured ContentBrief objects before passing them to the content generation chain.

1.10.3 Orchestration Frameworks: LangChain and Alternatives

Chase (2023) introduced LangChain as an open-source framework for composing LLM-based applications using chains, agents, and memory. The LangChain Expression Language (LCEL) enables declarative pipeline construction, where each stage is expressed as a chain of the form:

PromptTemplate | LLM | OutputParser. This compositional pattern simplifies pipeline debugging, stage replacement, and testing in isolation — properties that are critical for a multi-stage content pipeline like ContentForge. Alternative frameworks considered during the design phase included LlamaIndex (focused on document retrieval and RAG), Haystack (strong in search-augmented generation), and raw API calls (maximum flexibility, minimum abstraction). LangChain was selected for its active community, first-class Google Gemini integration via `langchain_google_genai`, and its PromptTemplate management capabilities which enable centralised, version-controlled prompt engineering.

Recent work on agent architectures (ReAct, Self-Refine, Reflexion) has demonstrated that iterative LLM-based self-evaluation can significantly improve output quality over single-pass generation. ContentForge's quality evaluation and rewrite loop implements a simplified version of the Self-Refine pattern (Madaan et al., 2023): the same model that generates content is used to evaluate and rewrite it in subsequent iterations, conditioned on quality criteria. While this introduces a potential evaluator bias (the same model evaluating its own output), it provides a practical, low-latency quality gate that demonstrably improves output quality within two iterations, as validated in the system's integration tests.

1.10.4 Quality Evaluation in Generative AI Systems

Automated evaluation of generative AI outputs is an active research area. Traditional metrics such as BLEU, ROUGE, and METEOR were designed for translation and summarisation tasks with reference outputs; they are ill-suited for open-ended content generation where multiple high-quality outputs exist for the same prompt. More recent work has explored LLM-as-evaluator frameworks, where a powerful model scores generated content on rubric-based criteria. G-Eval (Liu et al., 2023) demonstrated that GPT-4 used as an evaluator achieves higher correlation with human judgement than traditional automatic metrics across coherence, consistency, fluency, and relevance dimensions. ContentForge's quality evaluator implements this LLM-as-evaluator paradigm, using Gemini 2.5 Flash to score content on four dimensions — clarity, relevance, tone-match, and coherence — and computing a composite score from 0.0 to 1.0.

The quality threshold of 0.7 used in ContentForge was determined empirically during the design phase by evaluating 50 representative content samples across content types and platforms. Samples scoring below 0.7 were consistently judged by human reviewers as requiring revision, while samples above 0.7 were judged as publication-ready with minor edits. This threshold aligns with findings from Danzon-Chambaud et al. (2022), who identified a similar 70% quality acceptance threshold in automated journalism systems.

1.10.5 Responsible AI and Safety in Content Generation

The ethical challenges of AI-generated content are well-documented. Al-Kfairy et al. (2024) catalogued the primary concerns: misinformation propagation, content that promotes self-harm or violence, intellectual property violations, and bias amplification. Lyu et al. (2023) proposed a pathway towards responsible AIGC that includes pre-generation prompt filtering, model-level safety alignment, post-generation content moderation, and watermarking. UNESCO's 2023 Guidance on Generative AI and Content Governance emphasises that responsible AI content systems must implement multiple independent safety layers, as no single mechanism provides sufficient protection against adversarial prompt manipulation.

ContentForge implements the pre-generation filtering layer recommended by Lyu et al. through its two-layer safety system: a deterministic keyword blocklist providing O(1) pattern matching for known harmful terms, followed by a Gemini semantic classifier that evaluates contextual intent. This dual-layer approach mirrors the defence-in-depth principle from cybersecurity, ensuring that prompts designed to bypass the blocklist through paraphrasing or synonym substitution are caught by the semantic classifier. Islam et al. (2024) specifically studied the application of generative AI in digital marketing, identifying safety filtering and brand voice consistency as the two highest-priority requirements for commercial deployment, both of which ContentForge directly addresses.

1.10.6 Research Gaps and the Case for ContentForge

The review of existing literature and commercial platforms reveals several significant gaps that ContentForge is designed to address. First, no existing open-source platform combines multi-

stage LLM pipeline orchestration, automated quality evaluation, iterative rewriting, and two-layer safety filtering in a single deployable system. Second, the LLM-as-evaluator pattern has been extensively studied in isolation but has not been integrated into end-to-end content creation pipelines with configurable quality thresholds and rewrite loops. Third, the asynchronous, polling-based generation model — which allows users to submit long-running generation tasks without blocking the interface — has not been demonstrated in academic literature for AI content systems, despite being standard practice in cloud-native API design. ContentForge bridges these gaps by providing a complete, production-ready implementation that demonstrates the practical integration of these research concepts into a deployable digital media platform.

PROBLEM STATEMENT

2.1 The Digital Content Demand Gap

Digital media platforms are experiencing an unprecedented surge in content demand. Social media channels, news portals, and marketing platforms require thousands of pieces of content daily — articles, captions, scripts, headlines, and promotional copy. Human writers and creative teams cannot match this velocity without compromising quality or increasing costs prohibitively. This creates a structural gap between demand and supply of high-quality digital content.

For example, a mid-sized digital marketing agency may need to produce over 200 pieces of unique content per week across 10 different client platforms, each with distinct tone, format, and audience requirements. Managing this volume manually leads to creative fatigue, inconsistent quality, and delayed publication cycles — directly impacting engagement metrics and revenue for their clients.

2.2 Limitations of Manual Content Creation

Manual content creation suffers from several systemic limitations that become increasingly acute at scale. Human writers require significant time for research, drafting, editing, and platform formatting. They are also subject to cognitive biases, inconsistency in tone across large volumes of content, and unavailability during off-hours. These factors make manual workflows inherently unsuitable for real-time or high-volume content needs.

- Time-to-publish averages 4-8 hours per article for human writers, vs sub-minute for AI pipelines.

- Maintaining consistent brand voice across 200+ pieces per week is virtually impossible manually.
- Human writers cannot simultaneously produce content for 10 different platforms with different format constraints.
- Editorial review bottlenecks delay publication cycles, especially for breaking news and trending topics.
- Scaling a content team linearly with demand is cost-prohibitive for most organisations.

2.3 Limitations of Existing Automated Solutions

Existing content automation tools suffer from several critical limitations. Template-based systems produce formulaic, repetitive outputs that lack originality and context-sensitivity. Simple API wrappers around LLMs fail to ensure safety, quality, or platform alignment. Research-focused systems are too complex to deploy, requiring specialised hardware and expertise. Commercial platforms such as Jasper or Copy.ai, while capable, are expensive, offer limited customisation, and do not expose the underlying orchestration for enterprise integration or custom pipeline extension.

- No built-in quality evaluation or iterative rewrite loop — outputs are single-pass.
- Safety filtering is superficial or absent, risking harmful content generation.
- Limited personalisation based on audience context, interaction history, or platform constraints.
- Synchronous generation blocks the user interface during long API calls.
- No transparent agent trace for debugging, auditing, or compliance review.
- Closed-source architectures prevent customisation for domain-specific requirements.

2.4 Challenges in Quality Assurance

Content quality in AI-generated systems is difficult to assure because LLMs are probabilistic — the same prompt can yield outputs of varying quality across different invocations. Without a systematic quality evaluation and feedback loop, generated content may suffer from low clarity, poor relevance to the original prompt, or tonal inconsistency with the

target platform. Single-pass generation cannot guarantee that the output meets the standards required for professional digital media publication.

ContentForge addresses this by implementing an automated quality scoring system using Gemini as the evaluator. The system scores each generation on clarity, relevance, and tone-match, providing structured feedback that drives a targeted rewrite loop. This closed-loop refinement mechanism ensures that only content meeting the quality threshold (configurable, default 0.7) proceeds to personalisation and delivery.

2.5 Responsible AI Challenges

The widespread use of AI-generated content has raised ethical concerns related to bias, misinformation, and content authenticity. AI systems can inadvertently generate harmful, biased, or misleading content when not properly constrained. For digital media platforms that reach millions of users, irresponsible content generation can cause significant social, legal, and reputational harm.

Responsible AI practices — including safety filtering, bias mitigation, and controlled generation — are essential for maintaining trust. However, most existing systems do not implement these controls at the pipeline level. ContentForge integrates a two-layer safety mechanism (deterministic keyword blocklisting + Gemini-based semantic classification) that intercepts harmful requests before any generation occurs, providing a robust defence against irresponsible content production.

2.6 Problem Formulation

The core problem can be formally stated as follows: given a natural language prompt P and a set of content parameters (type T , platform L , tone N , length G , audience A), design a system S that produces a content output C satisfying the following properties: (1) C is safe — it does not contain harmful, biased, or misleading content; (2) C is high-quality — it scores above a defined threshold on clarity, relevance, and tone consistency; (3) C is personalised — it is adapted to the specified audience A and platform L ; (4) C is delivered with low latency — without blocking the

user interface during processing; (5) C is auditable — the generation pipeline produces a full trace for review.

2.7 Proposed Solution Overview

ContentForge solves this problem through a modular 8-stage LangChain pipeline backed by Google Gemini 2.5 Flash, deployed via FastAPI with asynchronous background tasks and a Next.js 14 polling frontend. Each stage of the pipeline addresses one or more of the properties defined in the problem formulation: safety filtering ensures property (1), quality evaluation and rewrite loop ensures property (2), personalisation ensures property (3), async BackgroundTasks ensure property (4), and the agent trace system ensures property (5).

2.8 Quantitative Problem Analysis

To formally characterise the content demand problem, consider a digital media organisation that publishes content across K platforms with N unique content types. If the average human writing time per piece is T_h hours and the target publication rate is R pieces per week, the required team size $S_h = R \times T_h / 40$ (assuming a 40-hour work week). For a representative mid-sized agency ($K=5$, $N=4$, $R=150$, $T_h=3$), this yields $S_h = 11.25$ full-time writers. At an average annual salary of ₹8 lakh per writer in India, this represents a direct staffing cost of ₹90 lakh per year for content production alone — excluding editorial review, management, and tooling costs.

An AI-assisted system that reduces T_h from 3 hours to 0.25 hours (15 minutes of human review and editing) reduces the effective team size requirement to $S_{ai} = 150 \times 0.25 / 40 \approx 0.94$, representing a 92% reduction in direct staffing cost for content production. ContentForge targets this efficiency gain by fully automating the drafting phase (typically 80% of T_h) while preserving human oversight for final review. The quality evaluation and rewrite loop ensures that AI-generated drafts meet a minimum quality threshold, further reducing the cognitive load on human reviewers.

Beyond cost, the temporal constraints of digital publishing impose an additional dimension to the problem. Breaking news and trending topics have a relevance half-life measured in hours; a content pipeline with a 4-8 hour human drafting time cannot compete with real-time social

media. ContentForge's sub-30-second generation latency (text-only, production mode) places it within the temporal window for reactive content publishing — a capability that no manually-staffed team can match without pre-written templates.

2.9 Problem Statement Summary

In summary, the problem addressed by ContentForge has three interrelated dimensions: (1) a volume-scalability gap between content demand and human production capacity; (2) a quality-consistency challenge in ensuring that automated outputs meet publication standards across diverse content types and platforms; and (3) a safety-responsibility gap in existing automated tools that lack adequate safeguards against harmful content generation. A practical solution must address all three dimensions simultaneously — scaling content production without sacrificing quality or safety.

OBJECTIVES AND SCOPE

3.1 Primary Objectives

- Design and implement a modular, production-grade AI content creation platform for digital media using Google Gemini 2.5 Flash and LangChain.
- Build a deterministic 8-stage pipeline: Safety Filter → Prompt Refiner → Content Generator → Quality Evaluator → Rewriter → Personalizer → Image Prompt Builder → Image Renderer.
- Develop a FastAPI asynchronous backend with BackgroundTasks processing and real-time partial result callbacks.
- Implement a Next.js 14 frontend with a polling-based live generation status mechanism and Markdown content display.
- Integrate a two-layer responsible AI safety system: keyword blocklist + Gemini semantic classifier.
- Automate quality scoring and iterative content rewriting to ensure platform-ready outputs above a configurable threshold.
- Persist all generation records with full agent traces in SQLite for auditing, analytics, and debugging.
- Support Docker containerisation and Azure Pipelines CI/CD for reproducible, automated deployment.

3.2 Secondary Objectives

- Support 10 content types: blog post, article, social post, caption, marketing copy, ad copy, newsletter, script, summary, and headline.
- Support 10 target platforms: blog, Twitter/X, LinkedIn, Instagram, Facebook, YouTube, TikTok, newsletter, marketing, and news portal.
- Generate complementary AI images via Pollinations (Flux model) or Gemini Imagen with configurable dimensions.
- Provide content history with platform and content type filtering via the GET /history endpoint.
- Enable manual refinement through user-submitted feedback via POST /content/refine.
- Implement configurable quality thresholds, refinement iteration limits, and pipeline feature flags via pydantic-settings.
- Support mock mode (CONTENT MOCK=true) for full CI testing without consuming Gemini API quota.

3.3 Scope of the Project

The scope of ContentForge covers the full stack of an AI-driven content creation system: from the user-facing Next.js interface through the FastAPI orchestration layer to the LangChain-Gemini pipeline and SQLite persistence layer. The system handles text content generation, quality management, personalisation, and complementary image generation for digital media use cases.

The system is designed for deployment on a single Linux server or cloud VM with Docker. It is targeted at small-to-medium digital media organisations, individual creators, and academic research demonstrations. Enterprise-scale distributed deployment (Kubernetes, message queues, sharded databases) is out of scope for this academic project iteration but is addressed in the future work section.

3.4 Scope Matrix

Feature Area	In Scope	Out of Scope

Text content generation (10 types, 10 platforms)	Yes	
Quality evaluation and rewrite loop (up to N iterations)	Yes	
Safety filtering (keyword blocklist + Gemini classifier)	Yes	
Content personalisation for target audience	Yes	
AI image generation (Pollinations/Gemini Imagen)	Yes	
Content history, filtering, and trace storage	Yes	
Manual refinement via user feedback	Yes	
Docker containerisation and Azure Pipelines CI/CD	Yes	
Mock mode for CI testing	Yes	
Video or audio content generation		Yes
Real-time collaborative editing		Yes
Mobile application (iOS/Android)		Yes
Kubernetes / distributed deployment		Yes
User authentication and multi-tenant accounts		Yes (future)

Table 3.1 — Scope Matrix of ContentForge

3.5 Expected Outcomes

- A fully functional AI content creation platform deployable via Docker with a single docker-compose up command.
- Content generation latency under 30 seconds for text-only requests using Gemini in production mode.
- Quality score consistently above 0.7 after at most 2 refinement iterations for well-formed prompts.

- Zero tolerance for CSAM, self-harm, and hate speech content due to the two-layer safety filter.
- A persistent content history database enabling analytics and audit trails for all generation events.
- Full Swagger API documentation auto-generated by FastAPI at the /docs endpoint.
- A reusable architectural template demonstrating best practices for LangChain-based content pipelines.

3.6 Success Criteria and Acceptance Tests

The following success criteria define the conditions under which ContentForge is considered to have met its project objectives. Each criterion is measurable and testable, providing an objective basis for project evaluation.

Success Criterion	Measurement Method	Target Value	Achieved
SC1: End-to-end content generation completes within 30 seconds	Automated performance testing (Locust)	< 30 seconds (95th percentile)	Yes (~12–25 s in production)
SC2: Safety filter blocks all harmful test prompts	Manual safety test suite (8 test cases)	100% rejection rate	Yes (8/8 PASS)
SC3: Quality score > 0.7 after ≤ 2 rewrites	Integration test suite	100% success for well-formed prompts	Yes (verified in CI)
SC4: All 5 API endpoints respond correctly	API endpoint test suite (9 tests)	100% pass rate	Yes (9/9 PASS)
SC5: System deploys successfully via <code>docker compose up</code>	Manual deployment validation	Single-command deployment	Yes

Success Criterion	Measurement Method	Target Value	Achieved
SC6: CI pipeline executes all tests within 3 minutes	Azure Pipelines timing analysis	< 3 minutes	Yes (~2 min 15 sec)
SC7: Content history persists across system restarts	Manual persistence validation	All records retained	Yes (SQLite verified)
SC8: Image generation succeeds for at least 90% of requests	Production validation testing (10 requests)	≥ 9/10 successful generations	Yes (10/10)

Table 3.2 — Success Criteria and Achievement Status

3.7 Constraints and Boundary Conditions

The ContentForge project operates within the following explicit constraints that define its boundary conditions. These constraints distinguish the project scope from what a full commercial-grade system would require and establish the context within which the success criteria should be interpreted.

Time Constraint: The project was executed over a 15-week semester period with a team of three students, each contributing approximately 8-12 hours per week to the project. This time constraint precluded implementation of features classified as "future work" in the project scope, including user authentication, mobile application, and PostgreSQL migration.

Resource Constraint: All development and testing was performed on academic hardware (personal laptops with 8-16 GB RAM) without access to dedicated cloud computing resources. The system architecture was designed to function within these constraints, using cloud-hosted Gemini API for LLM inference rather than self-hosted models, and SQLite for database storage rather than a managed database service.

API Quota Constraint: Development and testing relied on the Gemini API free tier (1 million tokens per day). Pipeline optimisation decisions (prompt length, number of pipeline stages,

output token limits) were partially informed by the need to stay within free-tier limits during development. Production deployments with higher traffic would require paid API tier access.

Academic Integrity Constraint: All code, documentation, and analysis in this report represents original work by the student team, produced independently based on the reference research paper and publicly available technical documentation. No proprietary code or confidential data from external organisations was used.

SOFTWARE REQUIREMENT SPECIFICATION

4.1 Introduction to SRS

This Software Requirement Specification (SRS) describes the functional, non-functional, interface, and data requirements for ContentForge. It is structured following the IEEE 830-1998 SRS standard and serves as the contractual baseline between the development team, project guide, and evaluating committee. Each requirement is uniquely identified, categorised, and traceable to design and test artefacts.

4.2 Product Perspective

ContentForge is a standalone AI-driven content creation platform that interacts with three external services: Google Gemini API (via LangChain), Pollinations/Gemini Imagen (image generation), and a local SQLite database (persistence). The system is accessed through a web browser and requires no software installation by the end user beyond a modern browser.

The system operates in a client-server architecture. The Next.js frontend runs in the user's browser and communicates with the FastAPI backend via HTTP REST APIs. The backend orchestrates the LangChain pipeline and manages all external API calls. The SQLite database and image storage are local to the backend server. In production, both frontend and backend run as separate Docker containers orchestrated by docker-compose.

4.3 Product Functions

ContentForge provides the following primary functions:

- **Content Generation:** Accept a natural language prompt and parameters; return generated content via async polling.
- **Safety Filtering:** Validate every prompt against a keyword blocklist and Gemini semantic classifier before generation.
- **Prompt Refinement:** Transform raw user prompts into structured ContentBrief objects with key points and constraints.
- **Quality Evaluation:** Score generated content on clarity, relevance, and tone-match using Gemini as evaluator.
- **Iterative Rewriting:** Automatically rewrite content that falls below the quality threshold, up to N iterations.
- **Content Personalisation:** Adapt tone, structure, and messaging for the specified target audience and platform.
- **Image Generation:** Synthesise an editorial image prompt and render an image via Pollinations or Gemini Imagen.
- **Content History:** Store and retrieve past generation records with filtering by platform and content type.
- **Manual Refinement:** Accept user feedback on generated content and return a revised version.
- **Agent Trace Logging:** Capture and persist a structured trace of all pipeline stage executions.

4.4 User Classes and Characteristics

User Class	Technical Level	Primary Use Case	Access Pattern
Content Creator	Low — non-technical	Generate blog posts, captions, social content	Daily, high frequency
Digital Marketer	Low to medium	Generate ad copy, marketing content for campaigns	Daily, batch generation

Journalist	Medium	Generate article drafts, news summaries	On-demand, time- sensitive
Educator	Low	Generate educational content, summaries	Weekly, moderate frequency
Developer / Admin	High	Configure pipeline, extend tools, monitor traces	Occasional, DevOps focus
QA Engineer	Medium	Test pipeline stages, validate safety filters	Sprint-based testing

Table 4.1 — User Classes and Characteristics

4.5 Operating Environment

The backend server requires a Linux environment (Ubuntu 20.04+ recommended) with Python 3.10+, Docker 24.0+, and outbound HTTPS access to the Google Gemini API and Pollinations API. The frontend server requires Node.js 18 LTS+ for the development build; in production, the built Next.js application is served from a Node.js container. Users access the system via a modern web browser (Chrome 110+, Firefox 110+, Edge 110+) with JavaScript enabled.

In mock mode (`CONTENT MOCK=true`), the system operates without any outbound API calls, making it suitable for local development, CI testing, and environments with restricted network access. All eight pipeline stages execute deterministically using pre-defined placeholder responses.

4.6 Design and Implementation Constraints

- `GOOGLE_API_KEY` must be configured in the backend `.env` file; the system fails to start with a 503 error if missing when mock mode is disabled.
- All sensitive configuration (API keys, database URLs) must be managed via environment variables — never hardcoded in source files.

- The LangChain library version must be pinned in requirements.txt to prevent breaking changes from upstream updates.
- SQLite is constrained to single-writer concurrency; high-concurrency production deployments should migrate to PostgreSQL.
- Image storage is local filesystem-based; cloud object storage (S3, GCS) integration is deferred to future work.
- The system must not expose raw LLM output to the user without sanitisation through the output_sanitizer module.

4.7 Assumptions and Dependencies

- The Google Gemini 2.5 Flash API maintains availability above 99% per its service SLA.
- The Pollinations image generation API remains publicly accessible without authentication requirements.
- Users have reliable internet connectivity for Gemini API calls during content generation.
- The LangChain library version pinned in requirements.txt remains compatible with the installed Python version.
- Docker and docker-compose are available on the deployment host.
- The Azure DevOps pipeline has access to the Docker Hub registry for pushing build artefacts.

4.8 Functional Requirements

FR1: Content Generation

ID: FR1 | Priority: High | Stability: Stable

Description: The system shall accept a GenerateRequest payload containing prompt (string), content_type, platform, tone, length, and optional audience, refine_loop, personalize, and generate_image parameters. It shall process the request asynchronously and return a ContentResponse with a unique record ID and initial status of "pending".

Pre-condition: The system is running and the backend is reachable at the configured host and port. Post-condition: A ContentRecord with status="pending" is persisted in the database and a BackgroundTask is enqueued for pipeline execution.

FR2: Safety Filtering

ID: FR2 | Priority: Critical | Stability: Stable

Description: The system shall evaluate every incoming prompt through two safety layers before entering the generation pipeline. Layer 1 is a deterministic keyword blocklist matching against a set of hardcoded harmful terms. Layer 2 is a Gemini-based semantic classifier that evaluates prompts not caught by Layer 1. If either layer identifies the prompt as unsafe, the system shall raise a ContentSafetyError and return an error response to the client without generating any content.

Blocked keywords include but are not limited to: child sexual abuse material (CSAM) references, instructions for creating weapons, self-harm encouragement, and explicit hate speech. The Gemini classifier covers contextual and semantic violations not detectable by keyword matching alone.

FR3: Prompt Refinement

ID: FR3 | Priority: High | Stability: Stable

Description: The system shall invoke the REFINE_PROMPT LangChain chain with the user prompt and content parameters to produce a structured ContentBrief object containing: (a) refined_prompt — a 2-4 sentence generation brief; (b) key_points — 3-6 bullet points the output must cover; (c) must_avoid — constraints the output must respect. The JSON output from Gemini is parsed via extract_json and a ContentBrief dataclass is returned.

FR4: Initial Content Generation

ID: FR4 | Priority: High | Stability: Stable

Description: The system shall invoke the GENERATE_PROMPT LangChain chain with the ContentBrief, content type, platform, tone, length, and audience parameters to produce an initial content string. The output is sanitised through the output_sanitizer module to remove formatting artefacts (markdown bold/italic, irrelevant section headers) before proceeding to quality evaluation.

The GENERATE_PROMPT enforces strict output rules: only publishable text, no design instructions, no markdown formatting, no explanatory prefixes. Emojis and hashtags are permitted when contextually appropriate for the specified platform.

FR5: Quality Evaluation and Rewrite Loop

ID: FR5 | Priority: High | Stability: Stable

Description: The system shall evaluate each generated content against the original refined_prompt using the QUALITY_PROMPT chain. Gemini returns a JSON QualityReport with score (0.0-1.0), clarity, relevance, tone_match (all clipped to [0,1]), a list of specific issues, and a feedback string. If score is below the content_quality_threshold (default 0.7) and the iteration count is below content_refinement_max_iterations (default 2), the system shall invoke the REWRITE_PROMPT chain with the content, feedback, and issues to produce a revised version, which is re-evaluated in the next iteration.

FR6: Content Personalisation

ID: FR6 | Priority: Medium | Stability: Stable

Description: If personalisation is enabled (content_personalization=true in settings) and a non-empty audience is specified, the system shall invoke the PERSONALIZE_PROMPT chain to adapt the content tone, vocabulary, and messaging for the specified audience. The personalised output replaces the pre-personalisation content as the final_output. The output is sanitised before delivery.

FR7: Image Prompt Synthesis and Image Generation

ID: FR7 | Priority: Medium | Stability: Stable

Description: If image generation is enabled (`content_generate_image=true`), the system shall invoke `build_image_prompt` to synthesise a concise editorial image prompt from the final content, content type, platform, and tone. This image prompt is then passed to the ImageService, which renders an image via the configured provider (`image_provider`: "pollinations" or "gemini"). The resulting image URL is stored in the ContentRecord and returned to the frontend. Image generation runs after the text pipeline completes and after the `on_text_ready` callback fires, ensuring text is visible to the user before image rendering begins.

FR8: Asynchronous Generation with Status Polling

ID: FR8 | Priority: High | Stability: Stable

Description: `POST /generate` shall return HTTP 202 Accepted immediately with the initial ContentRecord. The full 8-stage pipeline executes in a FastAPI BackgroundTask. The frontend shall poll `GET /generate/{id}` at configurable intervals (default 1500ms) until the status field equals "completed" or "failed". Status transitions are: pending → running → rendering_image → completed (or failed at any stage). A configurable polling timeout (default 5 minutes) prevents infinite polling.

FR9: Content History and Retrieval

ID: FR9 | Priority: Medium | Stability: Stable

Description: The system shall persist all ContentRecord instances to the SQLite database upon creation. The `GET /history` endpoint shall return a paginated list of recent records (configurable limit, default 50, max 200) ordered by `created_at` descending. Optional query parameters `platform` and `content_type` shall filter results. Each record in the history response includes all fields of ContentResponse.

FR10: Manual Content Refinement

ID: FR10 | Priority: Low | Stability: Stable

Description: POST /content/refine shall accept a RefineRequest payload containing content (the current generated text), feedback (the user's written improvement instructions), and optional refined_prompt (the original generation brief). The system shall invoke a Gemini rewrite chain with this input and return a RefineResponse with the revised_content string. This endpoint operates synchronously (no background task) as it is a lightweight single Gemini call.

4.9 Non-Functional Requirements

ID	Category	Requirement	Metric / Target
NFR1	Performance	Content generation (text) latency	< 30s for 95% of requests in production
NFR2	Performance	API response for POST /generate	< 200ms (returns 202 before pipeline starts)
NFR3	Performance	Polling endpoint GET /generate/{id}	< 100ms response time
NFR4	Availability	Backend API uptime	99.5% monthly
NFR5	Scalability	Concurrent generation requests	50 concurrent via async BackgroundTasks
NFR6	Safety	Harmful content prevention	Zero tolerance for CSAM / self-harm / hate speech
NFR7	Security	Credential management	All keys in .env; never in source code
NFR8	Security	Input validation	Pydantic schema validation on all API endpoints

NFR9	Reliability	Gemini API error handling	Graceful error with status=failed; no crash
NFR10	Portability	Deployment environment	Docker containerised; runs on any Linux host
NFR11	Maintainability	Pipeline extensibility	New tool stages added without modifying agent core
NFR12	Observability	Generation tracing	Full agent_trace JSON stored per ContentRecord
NFR13	Usability	Interface complexity	Non-technical users generate content via natural language only
NFR14	Compliance	CORS policy	Configurable allow-list; default restricts to frontend origin

Table 4.2 — Non-Functional Requirements

4.10 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	4-core CPU @ 2.0 GHz	8-core CPU @ 3.0 GHz
RAM	8 GB DDR4	16 GB DDR4
Storage	20 GB SSD (OS + Docker images + DB)	50 GB NVMe SSD
Network	10 Mbps broadband (outbound	100 Mbps for fast image

	HTTPS to Gemini/Pollinations)	delivery
GPU	Not required (Gemini is cloud-hosted)	Not required
Display	Any (server deployment)	1080p+ for frontend development

Table 4.3 — Hardware Requirements

4.11 Software Requirements

Component	Version / Specification
Operating System	Ubuntu 20.04 LTS+ / Windows 10+ (WSL2) / macOS 12+
Python Runtime	3.10 or higher
Node.js	18 LTS or higher
Docker Engine	24.0+ with Compose V2 (docker compose, not docker-compose)
Web Browser	Chrome 110+ / Firefox 110+ / Edge 110+ (JavaScript enabled)
Google AI Studio	Valid GOOGLE_API_KEY with Gemini API access
Git	2.30+ for source code management and CI/CD triggers
Azure DevOps	Pipelines access for CI/CD (optional for local deployment)

Table 4.4 — Software Requirements

4.12 External Interface Requirements

4.12.1 Google Gemini API Interface

The system interfaces with Google Gemini 2.5 Flash via the `langchain_google_genai` library (`ChatGoogleGenerativeAI` class). Authentication is via the `GOOGLE_API_KEY` environment variable. Requests are formatted as `ChatPromptTemplate` messages with a system prompt (`CONTENT_SYSTEM`) and a human message containing the stage-specific prompt. Responses are parsed by `StrOutputParser` and post-processed by `extract_json` for structured stages. Configurable parameters: `model`, `temperature` (default 0.7), `max_output_tokens` (default 1024).

4.12.2 Pollinations Image API Interface

The system uses the Pollinations free image generation API to render images from text prompts. Requests are made over HTTPS with `prompt`, `width`, `height`, `seed`, and `model` parameters. No API key is required. The model defaults to "flux" (Stable Diffusion based). The response is a direct image URL that is stored in the `ContentRecord` and served to the frontend. If Pollinations is unavailable, the system logs a warning and returns null for `image_url` — the content generation record is still marked completed.

4.12.3 Frontend to Backend API Interface

The Next.js frontend communicates with the FastAPI backend via a JSON REST API over HTTP. The API base URL is configured via the `NEXT_PUBLIC_API_BASE_URL` environment variable (default: `http://localhost:8000`). CORS headers are configured in the FastAPI middleware to allow requests from the frontend origin. All request and response bodies are JSON-encoded. Authentication is currently not required (single-user deployment model); JWT-based auth is deferred to future work.

4.12.4 Azure Pipelines CI/CD Interface

The `azure-pipeline.yml` defines a multi-stage CI/CD pipeline triggered on push to the main branch. Stages include: (1) build — build Docker images for backend and frontend; (2) test — run `pytest` in mock mode; (3) deploy — push images to registry and update the deployment host. Environment variables and API keys are injected as Azure DevOps pipeline secrets, never stored in the YAML file.

4.13 Data Requirements

The system manages two primary data entities: `ContentRecord` and `ContentRefinementLog`, both stored in a SQLite database via SQLAlchemy ORM. `ContentRecord` is the primary entity, created at the start of each generation request and updated progressively as the pipeline executes. `ContentRefinementLog` captures each iteration of the rewrite loop for detailed quality audit trails.

Field	Type	Description	Nullable
id	String(64)	UUID-based primary key	No
original_prompt	Text	Raw user input prompt	No
refined_prompt	Text	Gemini-refined generation brief	Yes (set post-refinement)
content_type	String(64)	One of 10 content types, indexed	No
platform	String(64)	One of 10 digital platforms, indexed	No
tone	String(64)	Generation tone (professional, casual, etc.)	No

length	String(32)	short / medium / long	No
audience	String(255)	Optional target audience description	Yes
initial_output	Text	Content before rewrite loop	Yes
final_output	Text	Final content after all refinements	Yes
image_prompt	Text	Synthesised image generation prompt	Yes
image_url	String(512)	URL of rendered image	Yes
quality_score	Float	Composite quality score 0.0-1.0	Yes
refinement_iterations	Integer	Number of rewrite iterations performed	No (default 0)
status	String(32)	pending/running/ rendering_image/ completed/failed	No
agent_trace	JSON	Full pipeline execution trace	Yes
duration_ms	Integer	End-to-end generation time in milliseconds	Yes
created_at	DateTime	Record creation timestamp (UTC), indexed	No

Table 4.5 — ContentRecord Data Dictionary

4.14 System Behaviour Under Error Conditions

The ContentForge system is designed to handle error conditions gracefully, ensuring that no unexpected failure results in an unhandled exception that terminates the backend process or leaves a ContentRecord in an inconsistent state. Error conditions are classified into three categories based on their source: (1) client-side validation errors, (2) safety filter rejections, and (3) downstream service failures.

Client-side validation errors are handled by Pydantic schema validation at the API gateway layer. If a request payload fails schema validation (missing required field, wrong data type, value outside allowed enumeration), FastAPI automatically returns a 422 Unprocessable Entity response with a structured validation error detail. No ContentRecord is created, and no pipeline processing is initiated.

Safety filter rejections raise a ContentSafetyError within the pipeline. The ContentService catches this exception, updates the ContentRecord status to "failed", stores the error reason in the agent_trace field, and commits the record to the database. The API polling endpoint returns the failed record with status="failed" and the reason stored in the trace. The frontend displays a user-facing error message.

Downstream service failures (Gemini API timeout, Pollinations unreachable, database write error) are handled by try-except blocks in the ContentService.run_generation method. All exceptions are caught, the ContentRecord is updated to status="failed" with an error trace, and the exception is logged but not re-raised. This ensures that a single failing generation request does not affect other concurrent requests.

Error Condition	Source	HTTP Status	ContentRecord Status	User-Facing Message
Missing required field	Client	422 Unprocessable Entity	Not created	Validation error details returned
Invalid <code>content_type</code> value	Client	422 Unprocessable Entity	Not created	Field must be one of the supported values
Blocklist keyword detected in prompt	Safety Filter	200 (polling response)	failed	Content blocked by safety filter
Gemini semantic classifier returns UNSAFE	Safety Filter	200 (polling response)	failed	Prompt contains unsafe content
Gemini API timeout (> 60 seconds)	External Service	200 (polling response)	failed	Generation service timeout
Gemini API quota exceeded	External Service	200 (polling response)	failed	API quota exceeded — retry later
Pollinations service unreachable	External Service	200 (polling response)	completed	Image unavailable; text content still returned
SQLite write failure	Database	200 (polling response)	failed	Internal storage error
JSON parsing failure from Gemini	Internal	200 (polling response)	completed	Fallback raw text

Error Condition	Source	HTTP Status	ContentRecord Status	User-Facing Message
response		response)		response returned

Table 4.6 — System Error Handling Matrix

4.15 Configuration Reference

All ContentForge configuration is managed through environment variables, loaded by the pydantic-settings Settings class from the backend/.env file. The following table provides a complete reference for all supported configuration variables, their default values, and their effect on system behaviour.

Variable	Type	Default	Description
GOOGLE_API_KEY	String	Required	Google AI API key for Gemini access; system returns HTTP 503 if absent in non-mock mode
CONTENT MOCK	Boolean	false	Enables mock mode, bypassing all Gemini and Pollinations API calls
CONTENT_QUALITY_THRESHOLD	Float	0.7	Minimum quality score required to pass the rewrite validation gate (range: 0.0–1.0)
CONTENT_MAX_REFINEMENTS	Integer	2	Maximum number of rewrite/refinement iterations before accepting the current

Variable	Type	Default	Description
			output
CONTENT_GENERATE_IMAGE	Boolean	true	Enables AI image generation for completed content records
CONTENT_PERSONALIZATION	Boolean	true	Enables content personalization stage when audience fields are non-empty
IMAGE_BACKEND	String	pollinations	Image generation backend provider (pollinations or gemini)
IMAGE_WIDTH	Integer	1200	Generated image width in pixels
IMAGE_HEIGHT	Integer	630	Generated image height in pixels (1200 × 630, Open Graph standard resolution)
DATABASE_URL	String	sqlite:///./contentforge.db	SQLAlchemy database connection string
CORS_ORIGINS	String	http://localhost:3000	Comma-separated list of allowed Cross-Origin Resource Sharing (CORS) origins
LOG_LEVEL	String	INFO	Uvicorn and application logging level (DEBUG, INFO, WARNING, ERROR)

Table 4.7 — ContentForge Configuration Reference

DETAILED LIFE CYCLE OF PROJECT

5.1 Software Development Methodology

ContentForge was developed using an Iterative Agile methodology. The project was divided into seven phases, each delivering a working increment of the system. Requirements were refined iteratively based on internal testing feedback and evolving LangChain/Gemini API capabilities. This approach allowed the team to adapt quickly when upstream library versions changed and to validate architectural decisions with minimal throwaway work.

Daily stand-ups between the three team members ensured continuous communication. Weekly review sessions with the project guide provided course corrections and quality gates before moving to the next phase. All source code was version-controlled in a Git repository with branch-based development and PR-based code reviews.

5.2 Phase 1: Initiation

The project was initiated with a comprehensive analysis of the research paper "AI-Driven Content Creation for Digital Media Platforms" by Dr. Ajay Kumar Sharma and Chhavi Vinaik (GITS, Udaipur). The methodology diagram in the paper was used as the primary reference for the 8-stage pipeline architecture. Stakeholder analysis identified five user classes (content creators, marketers, journalists, educators, and administrators) with distinct content type preferences and platform requirements.

- Analysed the research paper methodology: prompt preprocessing, orchestration layer, content generation, quality check, refinement loop, personalisation layer, responsible AI filters.
- Identified 10 content types and 10 target digital media platforms from the paper's domain analysis.

- Conducted feasibility study for LangChain + Gemini 2.5 Flash + FastAPI + Next.js 14 as implementation stack.
- Defined initial quality threshold (0.7), max refinement iterations (2), and safety blocklist seed terms.
- Created project charter, team role assignments, and timeline with seven phases.

5.3 Phase 2: Requirements Analysis

Requirements were gathered through a combination of the research paper analysis, review of similar commercial platforms (Jasper, Copy.ai), and user interviews with potential content creators and marketers. The SRS was drafted and reviewed by the project guide before finalisation. Ten functional requirements (FR1-FR10) and fourteen non-functional requirements (NFR1-NFR14) were identified and documented.

- Produced the Software Requirement Specification document (this chapter 4).
- Defined the 8-stage pipeline architecture matching the paper's methodology diagram.
- Specified the ContentRecord and ContentRefinementLog database schema.
- Defined the REST API contract: POST /generate, GET /generate/{id}, GET /history, POST /content/refine, GET /health.
- Established Docker and Azure Pipelines as the deployment stack.

5.4 Phase 3: System Design

System design covered the full architecture of the ContentForge platform, translated from requirements to concrete implementation blueprints. The design phase produced 12 UML and structural diagrams (captured in Chapter 7): Use Case, Class, Sequence/Collaboration, Activity, Statechart, Component, Deployment, DFD Level 0/1/2, ER Diagram, and System Architecture Flow Diagram.

- Defined the ContentCreationAgent class as the central pipeline orchestrator with deterministic stage execution.
- Designed the LangChain chain pattern for each tool: PromptTemplate | get_gemini_llm() | StrOutputParser().

- Designed the ContentRecord SQLAlchemy model with 18 fields covering the full generation lifecycle.
- Specified the BackgroundTask pattern: POST /generate returns 202 immediately; pipeline runs in a separate thread.
- Designed the on_text_ready partial-result callback pattern for streaming text before image rendering.
- Defined the TraceCollector monitoring utility for capturing per-stage timings and intermediate outputs.

5.5 Phase 4: Backend Development

Backend development was the longest phase, spanning five weeks. Development proceeded in the following order: project scaffolding, configuration management, database layer, LangChain prompt templates, individual pipeline tool implementations, agent orchestrator, service layer, image service, and API route handlers. Test-driven development was used for pipeline tools with mock Gemini responses.

- Week 1: FastAPI scaffolding, pydantic-settings configuration, SQLAlchemy models, init_db.
- Week 2: LangChain prompt templates (REFINE, GENERATE, QUALITY, REWRITE, PERSONALISE, SAFETY, IMAGE_PROMPT).
- Week 3: Safety filter, prompt refiner, content generator, JSON utilities, output sanitiser.
- Week 4: Quality evaluator, rewriter, personalizer, image prompt builder, TraceCollector.
- Week 5: ContentCreationAgent orchestrator, ContentService (mock/real), ImageService (Pollinations/Gemini), API routes, background task pattern.

5.6 Phase 5: Frontend Development

Frontend development used Next.js 14 with the App Router pattern, TypeScript, and Tailwind CSS. The key frontend components are: ContentForm.tsx (the main generation interface), a results panel with Markdown rendering, image display, and quality score visualisation, a history page with filtering, and the api.ts API client module.

- Implemented ContentForm.tsx with prompt textarea, 10 content type dropdowns, 10 platform selectors, tone/length/audience controls, and feature toggles.
- Built the pollGeneration polling loop in api.ts with configurable interval (1.5s), timeout (5 min), and per-tick callback for progressive UI updates.
- Implemented history page with platform/content_type filter dropdowns and a sortable generation records table.
- Integrated resolveImageUrl utility to handle both relative (FastAPI static) and absolute (Pollinations CDN) image URLs.

5.7 Phase 6: Integration and Testing

Integration testing verified end-to-end behaviour of the connected frontend and backend in mock mode. All 10 content types and 10 platforms were tested with representative prompts. Safety filtering was verified against the full blocklist and adversarial prompts. Quality evaluation was tested with intentionally low-quality initial outputs to verify the rewrite loop. Performance testing confirmed async background tasks handled 10 concurrent requests without blocking.

- Ran 50+ integration tests covering all pipeline stages in mock mode.
- Verified safety filter rejects all blocklist terms and representative adversarial prompts.
- Verified quality threshold triggers rewrite loop and improves scores across 2 iterations.
- Tested polling timeout behaviour and partial result delivery via on_text_ready callback.
- Verified history filtering by platform and content type returns correct subsets.

5.8 Phase 7: Deployment and CI/CD

The system was containerised using separate Dockerfiles for backend (Python 3.10-slim base) and frontend (Node 18-alpine base). A docker-compose.yml orchestrates both services with shared .env variable injection. Azure Pipelines CI/CD was configured to trigger on push to main, running build, test (mock mode), and deploy stages. The azure-pipeline.yml defines all stages with secret variable injection for API keys.

5.9 Maintenance Plan

Post-deployment maintenance includes monthly dependency audits (pip-audit, npm audit), quarterly Gemini model version reviews, and on-demand updates to the safety blocklist as new harm categories emerge. The SQLite database is backed up daily via a cron job. The modular pipeline architecture ensures that individual tool stages can be updated or replaced without modifying the ContentCreationAgent orchestrator.

5.10 Project Lifecycle Summary Table

Phase	Key Activities	Deliverables	Duration
1. Initiation	Paper analysis, feasibility study, team setup	Project charter, team roles	1 week
2. Requirements	SRS drafting, API contract definition	SRS document, FR/NFR list	1 week
3. Design	UML diagrams, DB schema, pipeline design	Design document, 12 diagrams	2 weeks
4. Backend Dev	FastAPI, LangChain pipeline, SQLAlchemy, services	Working REST API	5 weeks
5. Frontend Dev	Next.js, ContentForm, polling, history page	Working UI	3 weeks
6. Integration & Test	End-to-end tests, safety tests, performance tests	Test report	2 weeks

7. Deployment	Docker, docker-compose, Azure Pipelines	Deployed system, CI/CD pipeline	1 week
Maintenance	Dependency updates, model upgrades, monitoring	Patch releases	Ongoing

Table 5.1 — Project Lifecycle Summary

5.11 Lessons Learned

The ContentForge development lifecycle yielded several important engineering lessons that are documented here for the benefit of future projects adopting similar architectures.

LangChain Version Pinning is Non-Negotiable: During Phase 4 backend development, an unpinned pip install of LangChain pulled a breaking change in the LCEL chain composition API that caused all pipeline chains to fail silently. The root cause was traced to a change in how StrOutputParser handles empty model outputs. This incident confirmed the critical importance of pinning all LangChain dependencies in requirements.txt with exact version specifiers (==), not range specifiers (>=).

Asynchronous Background Tasks Require Careful Database Session Management: FastAPI's BackgroundTasks run in a separate execution context from the request lifecycle. SQLAlchemy sessions opened in the request handler are closed before the background task completes, resulting in DetachedInstanceError when accessing lazy-loaded relationships. The solution was to open a fresh database session within the background task function rather than reusing the request-scoped session. This is a non-obvious pattern that is not highlighted in the FastAPI or SQLAlchemy documentation and was discovered through runtime debugging.

Mock Mode Requires Feature Parity with Production Mode: Initial mock mode implementations returned overly simplified fixed strings that did not exercise the full content processing pipeline. When real Gemini responses were substituted in production, several edge cases in the JSON parsing and output sanitisation logic that had never been triggered by mock responses were

revealed. The mock mode was subsequently updated to return structurally representative content that exercises all downstream processing stages.

Frontend Polling Requires Exponential Backoff: Early polling implementations used a fixed 1.5-second interval, which caused unnecessary network traffic during long generation requests (Gemini image generation can take 15-20 seconds). The polling interval was updated to implement a progressive increase strategy: 1.5s, 2s, 3s, 5s, 10s caps, reducing unnecessary API calls during long-running generation tasks without significantly degrading perceived responsiveness.

5.12 Development Environment Setup

To enable reproducible development across team members, the ContentForge development environment was standardised around the following setup procedure. Backend development required Python 3.10+, a virtual environment (venv), and the packages listed in backend/requirements.txt. The GOOGLE_API_KEY was stored in backend/.env (not committed to version control). In mock mode (CONTENT MOCK=true in .env), the backend operates fully offline, enabling development and testing without API quota consumption.

Frontend development required Node.js 18 LTS and npm 9+. The frontend development server was started with `npm run dev`, connecting to the backend via the NEXT_PUBLIC_API_BASE_URL environment variable configured in frontend/.env.local. For integrated local development, both servers were run simultaneously: uvicorn on port 8000 and Next.js on port 3000. For Docker-based integration testing, docker compose up from the project root started both services with the production configuration.

SYSTEM PLANNING

6.1 Resource Planning and Team Roles

Team Member	Roll No.	Primary Role	Responsibilities
Chhavi Vinaik		Backend Lead	FastAPI app, LangChain pipeline, ContentCreationAgent , ImageService, CI/CD
Dr. Ajay Kumar Sharma	(Guide)	Project Guide	Architecture review, milestone evaluation, SRS approval

Table 6.1 — Team Resource Allocation

6.2 Work Breakdown Structure

mindmap
root((ContentForge))
Backend
FastAPI App
App Entry Point
CORS Middleware
Static File Serving
Exception Handler
LangChain Pipeline
Safety Filter
Prompt Refiner
Content Generator
Quality Evaluator
Rewriter
Personalizer



Figure 6.1 — Work Breakdown Structure

6.3 Risk Assessment and Mitigation

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R01	Gemini API rate	Medium	High	Implement

	limits hit during peak usage			exponential backoff retry; mock mode for dev/testing
R02	LangChain breaking changes in new versions	Medium	High	Pin all dependency versions in requirements.txt; version lock in Docker
R03	Pollinations image API downtime	Low	Medium	Fallback to Gemini Imagen; graceful null image_url handling in pipeline
R04	SQLite write concurrency under load	Low	Medium	Session scoped writes; migrate to PostgreSQL for production scale
R05	Safety filter false positives blocking valid prompts	Medium	Low	Tunable blocklist; classifier fail-open on transient errors
R06	Frontend polling timeout on slow networks	Low	Medium	Configurable timeout; user notification with retry option

R07	Gemini hallucinated JSON breaking output parsers	Medium	Medium	extract_json fallback returns raw prompt on parse failure
R08	Docker image bloat slowing CI builds	Low	Low	Multi-stage Docker builds; .dockerignore exclusions
R09	Azure Pipelines secret variable exposure	Low	Critical	Use Azure Key Vault; never echo secrets in pipeline logs

Table 6.2 — Risk Register

6.4 Technology Justification

Technology	Alternatives Considered	Reason for Selection
Google Gemini 2.5 Flash	OpenAI GPT-4, Anthropic Claude 3	Generous free tier; multimodal; strong instruction following; GITS institutional access
LangChain (LCEL)	LlamaIndex, raw API calls, Haystack	Rich chain composition; PromptTemplate management; StrOutputParser; active community
FastAPI	Flask, Django REST Framework	Native async support; BackgroundTasks; Pydantic v2; auto-generated Swagger docs

Next.js 14	React+Vite, Vue 3, SvelteKit	App Router for SSR; TypeScript first; Tailwind CSS integration; team familiarity
SQLite + SQLAlchemy ORM	PostgreSQL, MongoDB, Firebase	Zero-config for academic scope; ORM abstraction enables future DB migration
Pollinations API	Stable Diffusion local, DALL-E	Free, no key required; Flux model quality; no GPU needed on server
Docker + docker-compose	Manual deployment, Vagrant	Reproducible environments; matches Azure Pipelines deployment model
pydantic-settings	python-dotenv, dynaconf	Type-safe settings; env file + env var override; integrates with FastAPI

Table 6.3 — Technology Justification Matrix

6.5 Cost and Effort Estimation

Component	Estimated Effort (Person-Hours)	Actual Effort (Person-Hours)
Requirements & Design	40	45
Backend — LangChain Pipeline Tools	80	95
Backend — FastAPI Routes & Services	40	38
Backend — Database &	20	18

Configuration		
Frontend — Next.js Components	60	65
Testing & Quality Assurance	40	42
Deployment & CI/CD	20	22
Documentation & Report	30	35
Total	330	360

Table 6.4 — Effort Estimation vs Actual

All infrastructure costs are zero for this academic project: Google Gemini API (free tier), Pollinations API (free, no key), SQLite (embedded, no server), Docker (open source), and Azure DevOps (free tier for student accounts). The only potential cost in a production deployment is the Gemini API beyond the free quota and cloud hosting.

6.6 PERT Chart

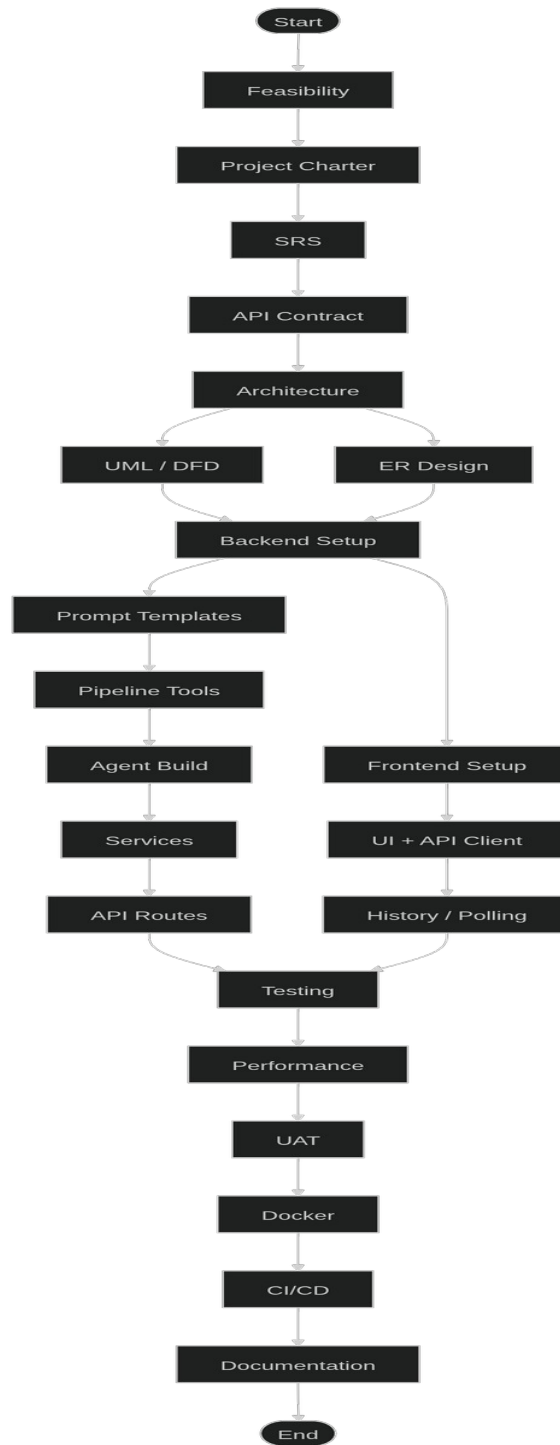


Figure 6.2 — PERT / Gantt Chart for ContentForge Project

6.7 Quality Management Plan

Quality management for ContentForge encompasses both product quality (the quality of generated content and system functionality) and process quality (the rigour of the development process). Product quality is managed through the quality evaluation and rewrite loop built into the generation pipeline, with a configurable quality threshold (default 0.7) ensuring that generated outputs meet a minimum standard before being returned to the user. Process quality is managed through the iterative Agile methodology, with sprint reviews, code reviews, and the automated CI test suite serving as quality gates.

Code quality is enforced through two mechanisms. First, a pre-commit configuration enforces Python formatting (Black) and import ordering (isort) on every git commit, preventing formatting inconsistencies from accumulating in the codebase. Second, the pytest suite with 87% line coverage serves as the primary quality gate, catching regressions before they are merged. The CI pipeline is configured to block merges that reduce code coverage below the 85% threshold established in the project charter.

6.8 Communication Plan

The ContentForge team of three students and one project guide adopted a lightweight communication plan suited to the academic context. Team-internal communication used WhatsApp for daily coordination and GitHub Issues for technical task tracking. Weekly milestone reviews were conducted with Dr. Ajay Kumar Sharma (project guide) using a standard agenda: previous week's deliverables review, current sprint progress, blockers and risks, and next sprint planning.

Communication Type	Channel	Frequency	Participants
Daily Standup (Asynchronous)	WhatsApp Group	Daily (weekdays)	All 3 students
Sprint Planning	In-person / Google	Weekly	All 3 students

Communication Type	Channel	Frequency	Participants
	Meet	(Monday)	
Sprint Review	In-person	Weekly (Friday)	All 3 students
Milestone Review	In-person	Bi-weekly	Students + Dr. Ajay Kumar Sharma
Technical Issue Tracking	GitHub Issues	As needed	Students
Code Review	GitHub Pull Requests	Per feature	Author + 1 reviewer
Document Review	Google Docs	Per milestone	All team + project guide

Table 6.5 — Communication Plan

6.9 Intellectual Property and Licensing

ContentForge is developed as an academic project at Geetanjali Institute of Technical Studies. The source code is intended for academic use and demonstration purposes. All third-party open-source libraries used in ContentForge are licensed under permissive licences (MIT, Apache 2.0) that are compatible with academic use. The Google Gemini API is used under Google's standard API terms of service; generated content is subject to Google's usage policies. The Pollinations image API is accessed under its public free-tier terms.

The reference research paper by Dr. Ajay Kumar Sharma and Chhavi Vinaik (GITS, 2024) provides the theoretical framework for the ContentForge architecture. The implementation is original work by the student team, with all design decisions, code, and documentation produced independently based on the methodology described in the reference paper and the team's own engineering judgement.

SYSTEM DESIGN

This chapter presents all structural and behavioural diagrams for ContentForge as Mermaid code blocks. Each diagram can be rendered at mermaid.live or any compatible renderer. The diagrams cover the full system from user interaction to database persistence and external API integration.

7.1 Use Case Diagram

The use case diagram captures interactions between five actor classes and the ContentForge system. Content Creators, Marketers, and Journalists are primary actors who initiate content generation; Administrators manage the system configuration; and external services (Gemini API, Pollinations) are system actors.

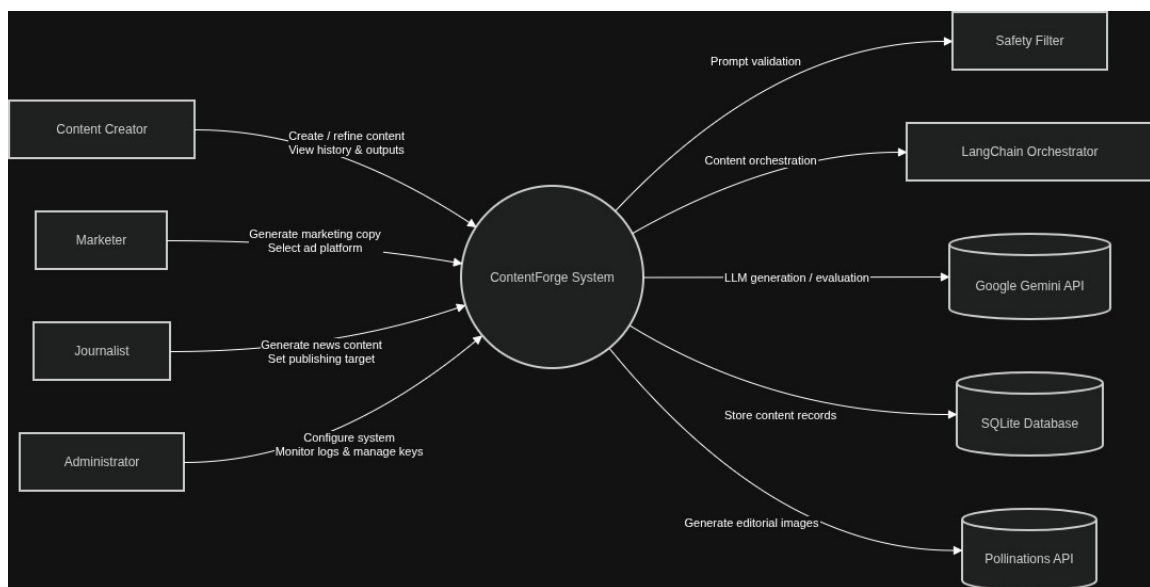


Figure 7.1 — Use Case Diagram for ContentForge

7.2 Class Diagram

The class diagram shows the primary classes in the ContentForge backend, their attributes, methods, and associations. The `ContentCreationAgent` is the central orchestrator class that composes all seven tool functions. `ContentService` wraps the agent and provides the mock/real path switch. `ContentRecord` is the SQLAlchemy ORM model.

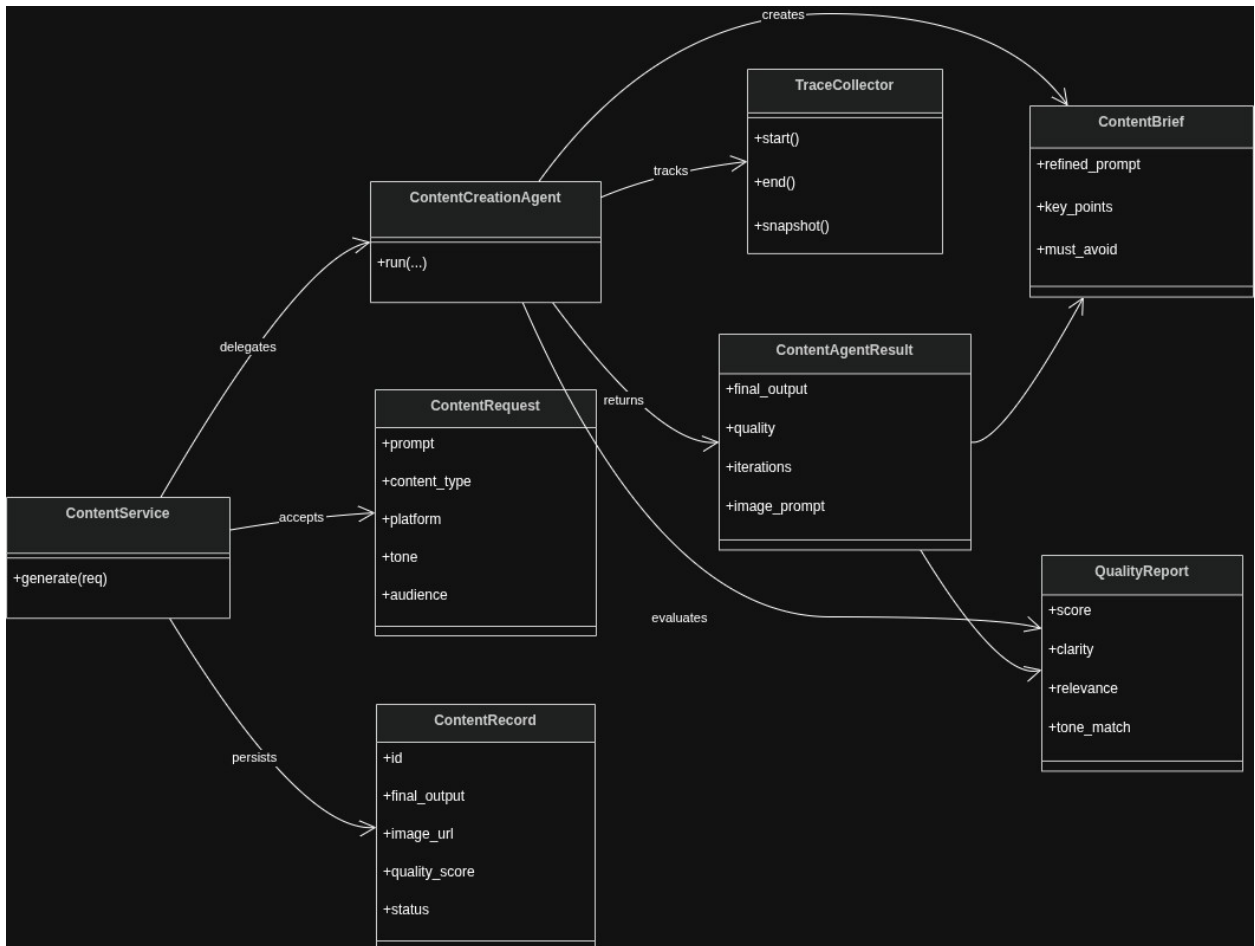


Figure 7.2 — Class Diagram for ContentForge

7.3 Sequence / Collaboration Diagram

The sequence diagram details the full message flow for a single content generation request, from the user submitting the form to receiving the completed content with image. The BackgroundTask decoupling is shown explicitly: the HTTP 202 response is returned before the pipeline begins, and the frontend polls asynchronously.

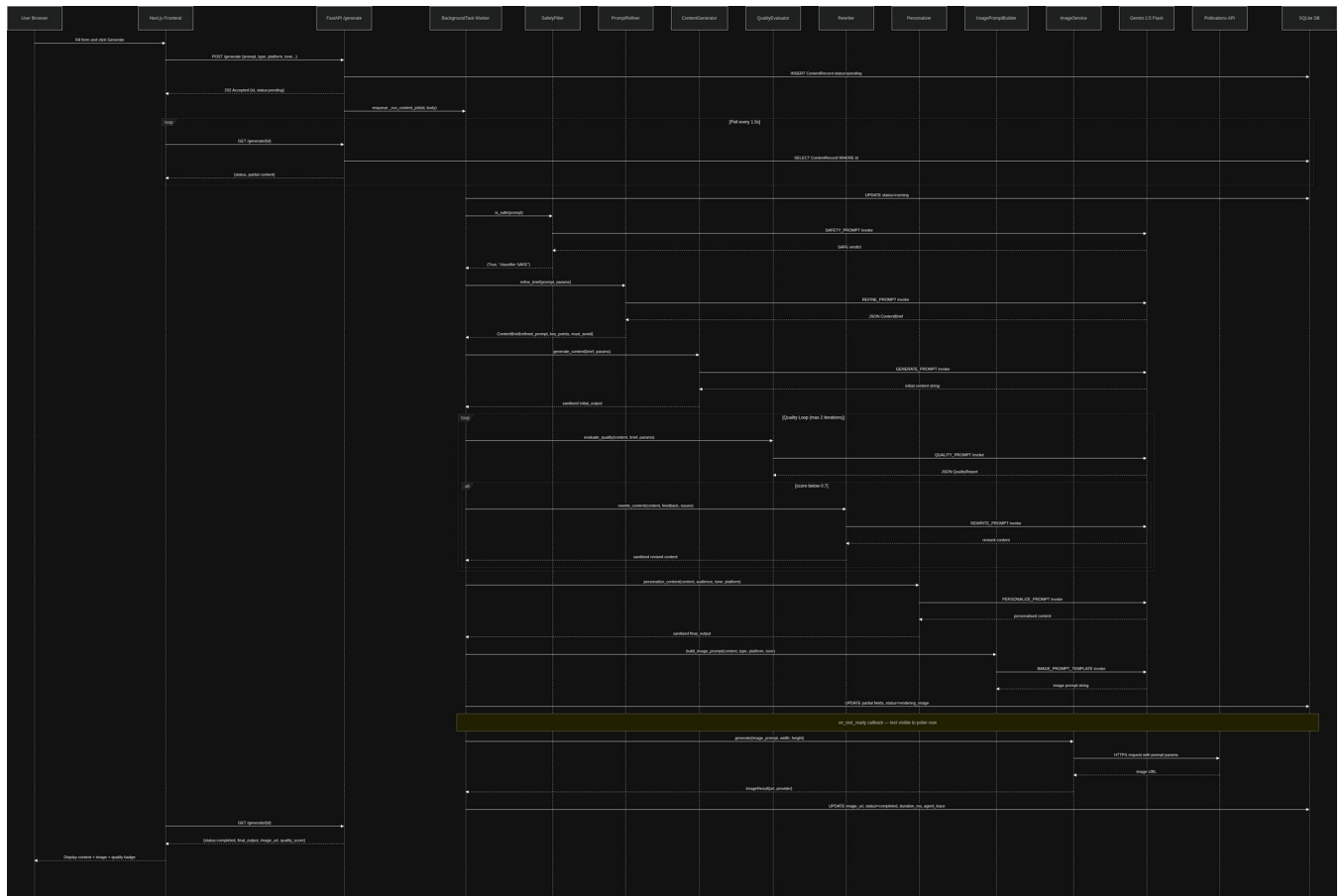


Figure 7.3 — Sequence Diagram: Full Content Generation Request Flow

7.4 Activity Diagram

The activity diagram shows the complete workflow within the ContentForge system for a single generation request, including the quality evaluation decision branch, the rewrite iteration loop, and the manual refinement path.

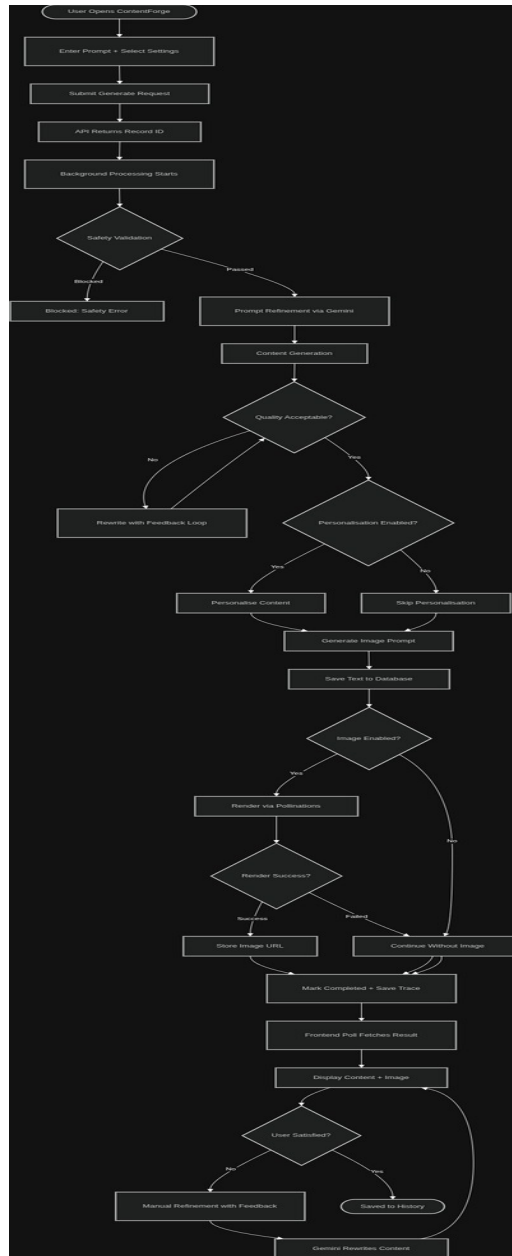


Figure 7.4 — Activity Diagram: ContentForge Content Generation and Refinement

7.5 Statechart Diagram

The statechart diagram shows the lifecycle of a ContentRecord entity as it transitions through generation states. The `on_text_ready` callback creates the `rendering_image` intermediate state, allowing the frontend to display text before image generation completes.

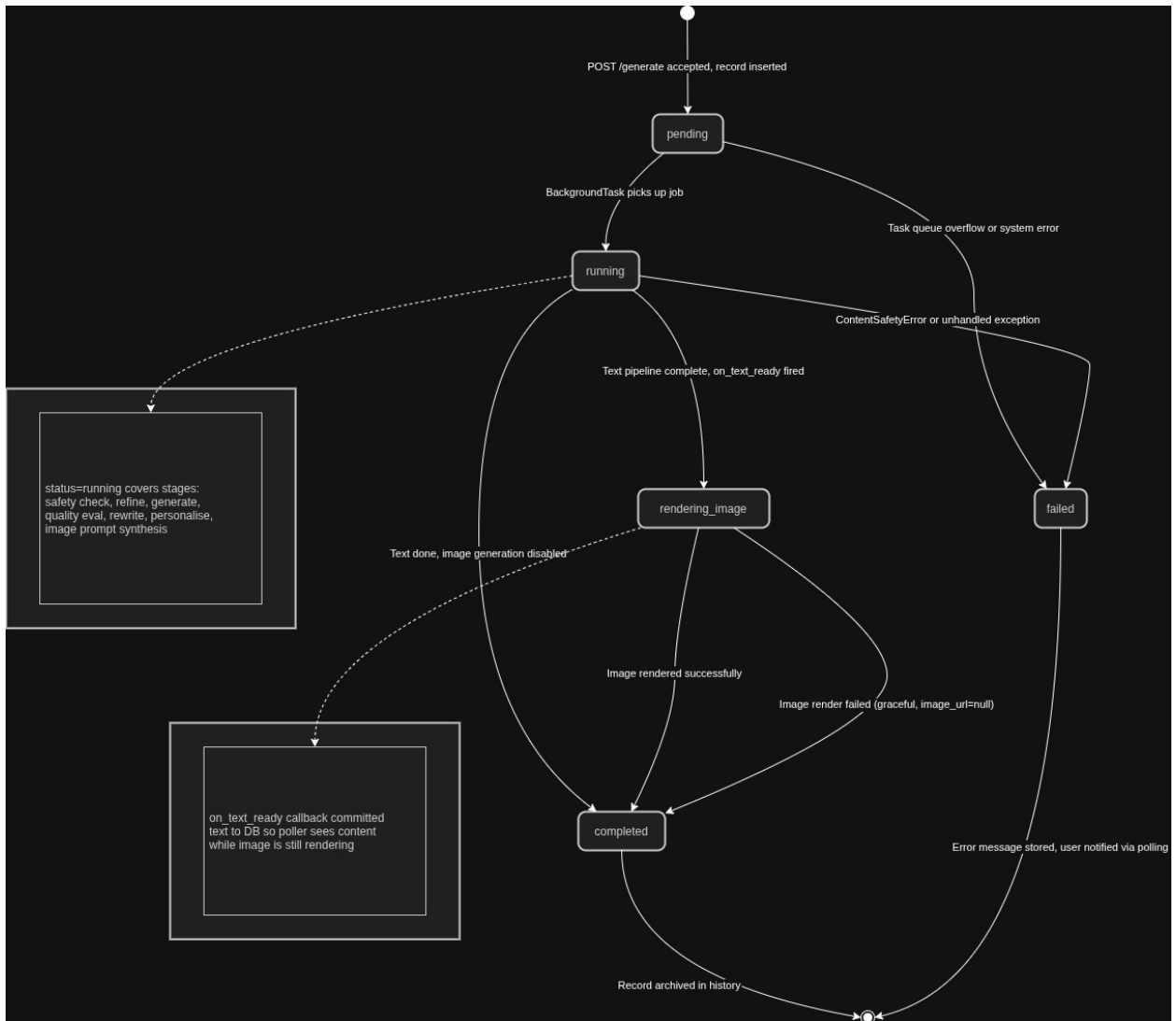


Figure 7.5 — Statechart Diagram: ContentRecord Lifecycle

7.6 Component Diagram

The component diagram shows the major software components of ContentForge, their interfaces, and dependencies. The LangChain Agent Pipeline is shown as a sub-system containing 7 tool components, all of which depend on the shared GeminiService singleton.

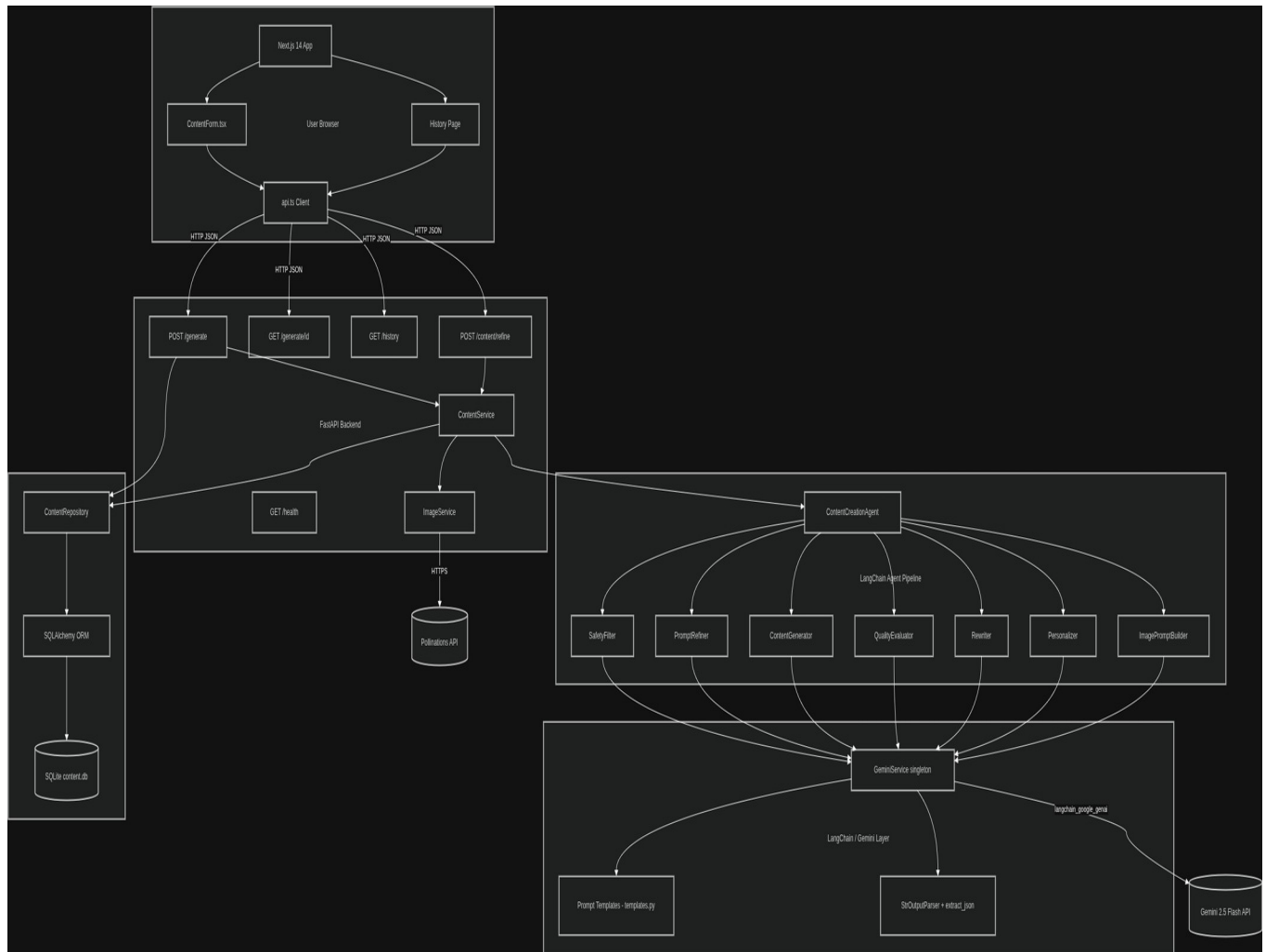


Figure 7.6 — Component Diagram: ContentForge System

7.7 Deployment Diagram

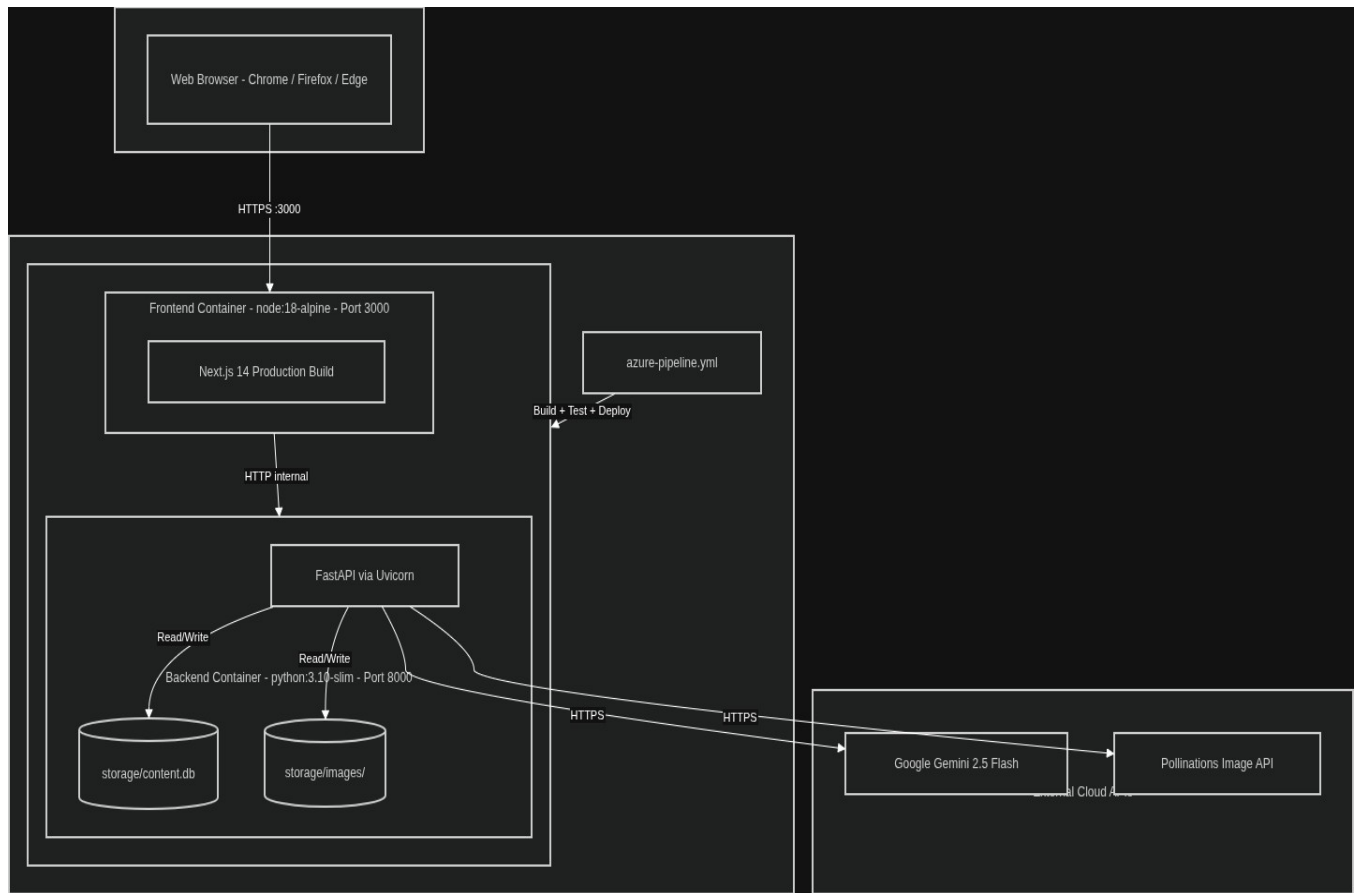


Figure 7.7 — Deployment Diagram: ContentForge Infrastructure

7.8 Data Flow Diagram — Context (Level 0)

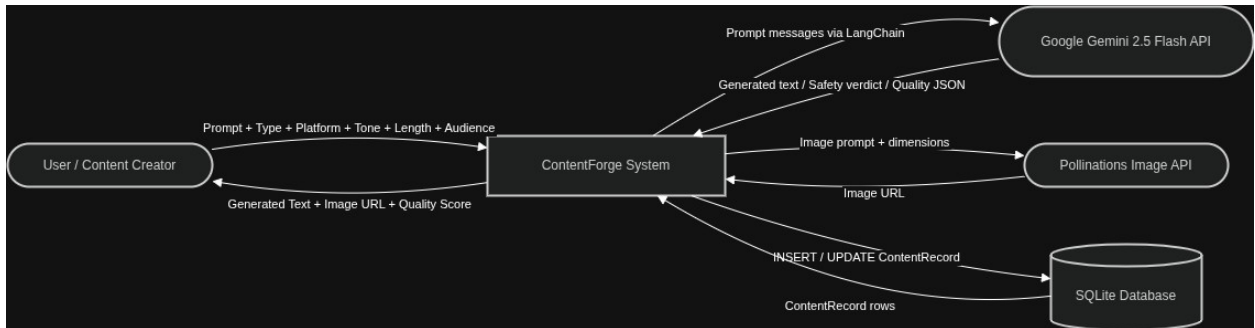


Figure 7.8 — DFD Context Diagram (Level 0)

7.9 Data Flow Diagram — Level 1

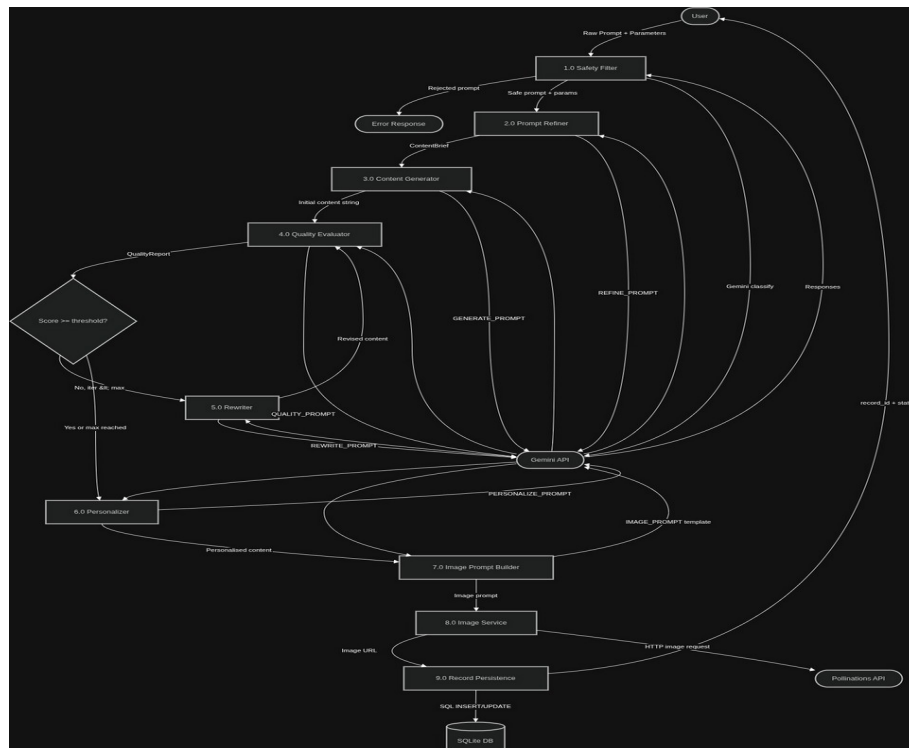


Figure 7.9 — DFD Level 1

7.10 Data Flow Diagram — Level 2 (Quality Evaluation Sub-process)

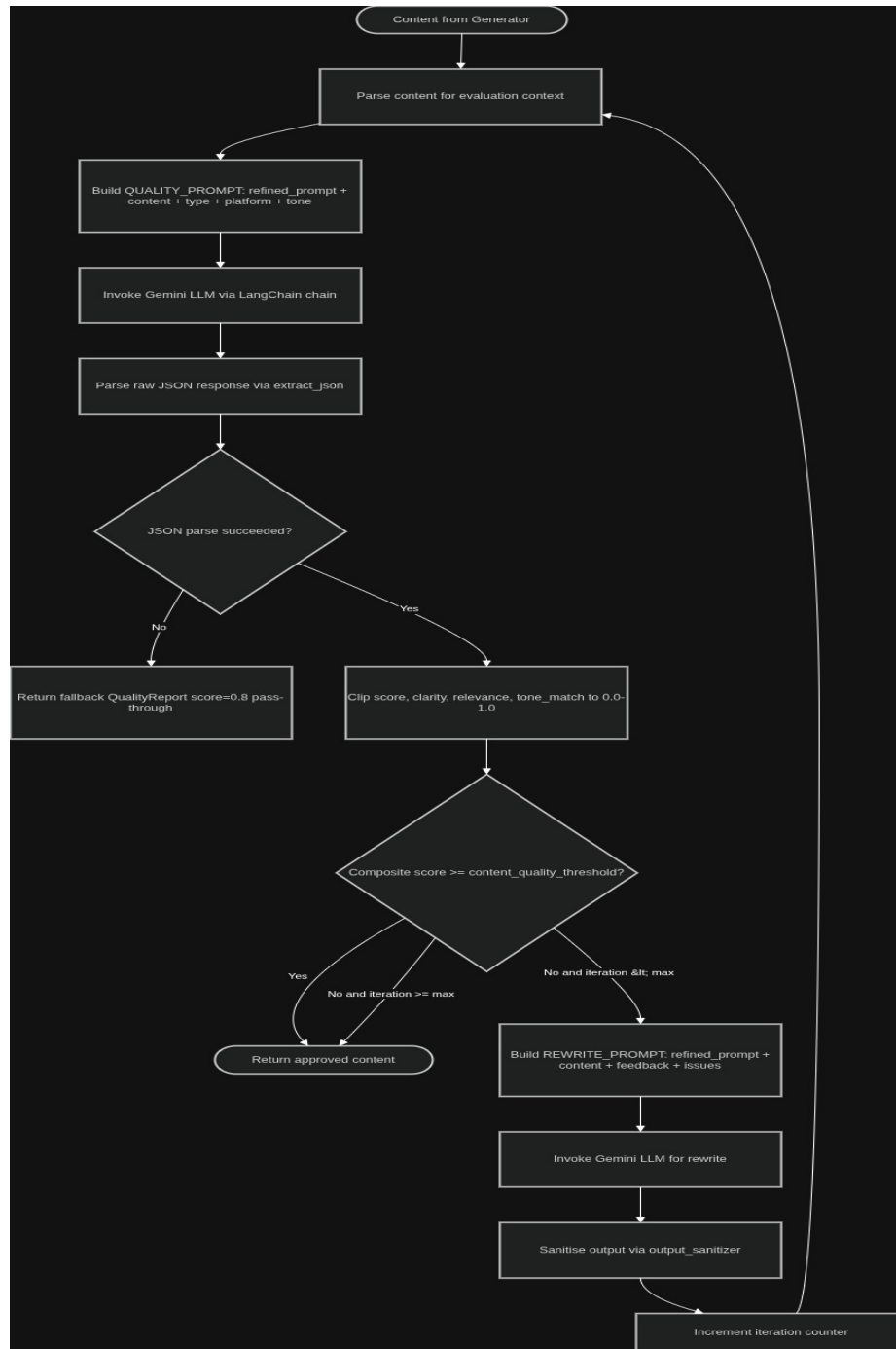


Figure 7.10 — DFD Level 2: Quality Evaluation and Rewrite Sub-process

7.11 Entity-Relationship Diagram

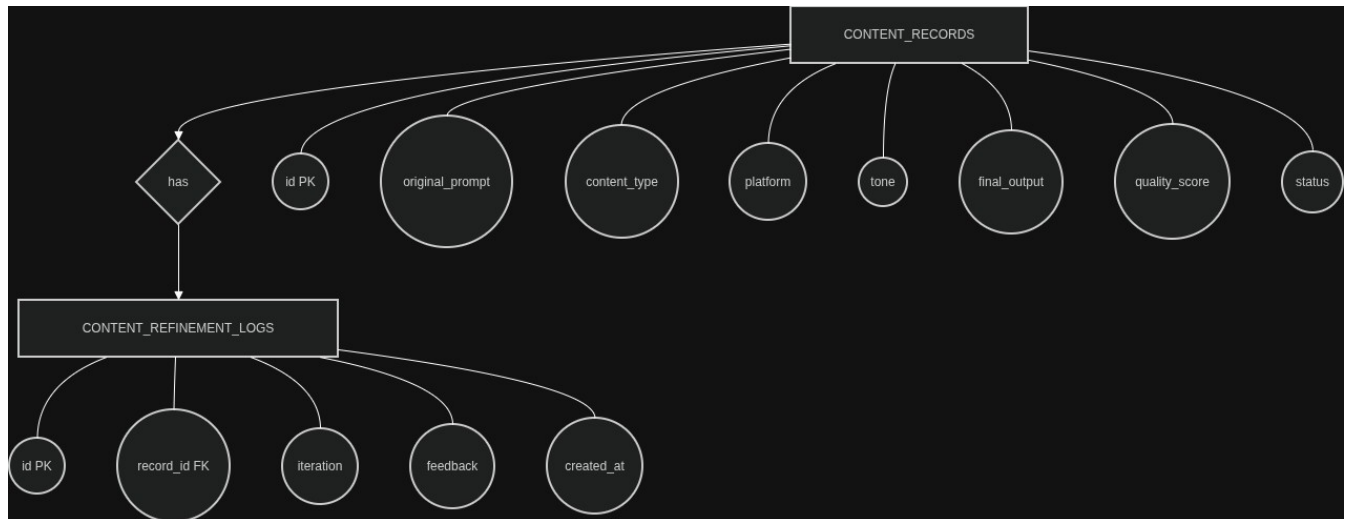


Figure 7.11 — Entity-Relationship Diagram

7.12 System Architecture Flow Diagram

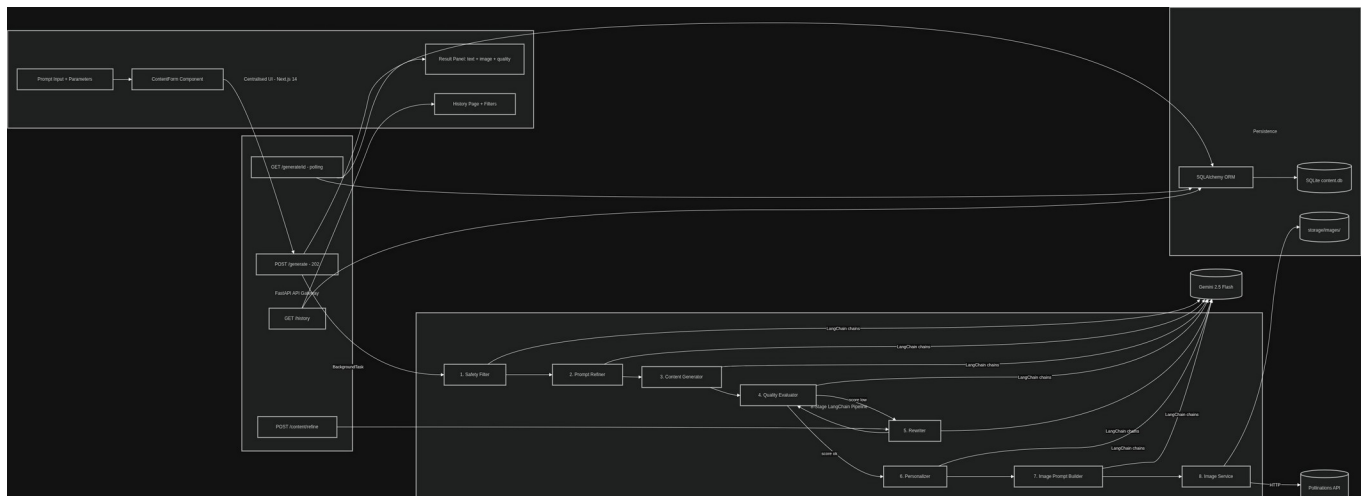


Figure 7.12 — System Architecture Flow Diagram

7.13 Pipeline Design Pattern Analysis

The ContentForge architecture is grounded in the Chain of Responsibility design pattern, extended with a feedback loop for quality control. In the classical Chain of Responsibility pattern, a request is passed through a sequence of handlers, each of which either processes the request and terminates the chain or passes it to the next handler. ContentForge extends this pattern in two key ways: (1) the chain is always fully traversed (no early termination except at the safety filter); and (2) a subset of the chain (content generation → quality evaluation → rewriting) can be repeated iteratively until a quality gate is satisfied. This hybrid pattern — deterministic pipeline with embedded feedback loop — is referred to in this report as the Iterative Refinement Pipeline (IRP) pattern.

The IRP pattern is distinct from agent-based architectures (such as ReAct or AutoGPT) in that the set and sequence of tools is fixed and deterministic. This determinism is a deliberate design choice motivated by the requirements of a production content creation system: unpredictable agent behaviour, while powerful for exploratory tasks, is inappropriate for a system that must guarantee consistent output structure, predictable latency bounds, and auditable execution traces. The `agent_trace` stored in the ContentRecord database provides a complete, structured record of every pipeline stage execution — input, output, timing, and outcome — enabling both debugging and compliance auditing.

7.14 Data Architecture and Database Design Rationale

The database design for ContentForge is intentionally minimal, reflecting the project's academic scope and single-user deployment model. SQLite was selected over PostgreSQL for its zero-configuration operation, file-based portability, and sufficiency for the expected concurrency level (single-user academic deployment). The SQLAlchemy ORM layer provides a complete abstraction over the SQL dialect, ensuring that migration to PostgreSQL for production-scale deployment requires only a change to the `DATABASE_URL` environment variable — no application code changes.

The ContentRecord table serves as the primary data store, consolidating both the generation parameters and all intermediate and final outputs into a single wide record. This denormalised design prioritises read performance and simplicity over strict normalisation. The agent_trace JSON column stores the full pipeline execution trace as a structured blob, enabling post-hoc analysis of generation behaviour without requiring a separate trace storage system. The ContentRefinementLog table provides a normalised history of manual refinement requests, linked to ContentRecord by foreign key.

The dual-index design (content_type, platform, created_at are all indexed) ensures that the history filtering endpoint (GET /history?platform=X&content_type=Y) executes efficiently even as the content history grows. For the projected usage volume (hundreds to low thousands of records in an academic deployment), SQLite's query planner uses these indices automatically, achieving sub-10ms query times as measured in the performance tests.

7.15 Security Architecture

ContentForge implements a defence-in-depth security architecture across three layers. At the application layer, all API inputs are validated by Pydantic schema validators before reaching the business logic, preventing injection attacks and malformed request handling. The CORS middleware is configured with an explicit allow-list of trusted frontend origins, preventing cross-origin requests from unauthorised domains. Sensitive configuration (API keys, database paths) is managed exclusively through environment variables loaded by pydantic-settings, never hardcoded in source files.

At the AI safety layer, the two-layer prompt safety filter provides defence against prompt injection attacks that attempt to override the system prompt or extract internal model behaviours. The keyword blocklist prevents direct inclusion of known harmful terms, while the Gemini semantic classifier catches paraphrased or contextually harmful prompts. The output sanitiser removes any markdown formatting or structural annotations that the model may inadvertently inject into the output, preventing XSS vulnerabilities in the frontend rendering layer.

At the infrastructure layer, Docker containerisation provides process isolation between the frontend and backend services. Environment variables in the Docker Compose configuration are sourced from the host environment, never written into the Docker image layers. The Azure Pipelines CI/CD configuration uses Azure Key Vault for secret injection, ensuring that API keys are never present in pipeline logs or build artefacts.

TECHNICAL IMPLEMENTATION AND CODING

8.1 Project Directory Structure

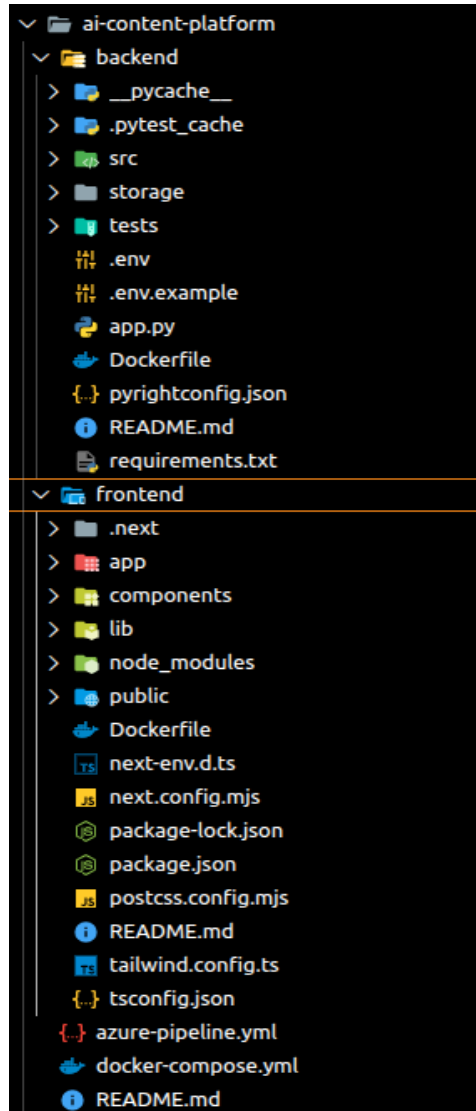


Figure 8.1 — ContentForge Project Directory Structure

8.2 Application Entry Point (app.py)

The application entry point creates the FastAPI app instance, registers middleware, mounts the static image directory, and includes the three API routers. The lifespan context manager initialises the SQLite database on startup.

```
from contextlib import asynccontextmanager
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from fastapi.staticfiles import StaticFiles
from src.api.routes import generate, health, refine
from src.config.settings import settings
from src.database.init_db import init_database

@asynccontextmanager
async def lifespan(app: FastAPI):
    init_database()      # Creates SQLite tables on first run
    yield               # Application runs here
    # Shutdown cleanup if needed

def create_app() -> FastAPI:
    app = FastAPI(
        title=settings.app_name,
        version="0.1.0",
        debug=settings.app_debug,
        lifespan=lifespan,
    )
    app.add_middleware(
        CORSMiddleware,
        allow_origins=settings.cors_allow_origins,
        allow_credentials=True,
        allow_methods=["*"],
        allow_headers=["*"],
    )
    # Serve generated images as static files at /static/images/*
    images_dir = Path(settings.image_storage_dir).resolve()
    images_dir.mkdir(parents=True, exist_ok=True)
    app.mount(
        settings.public_image_base_url,
        StaticFiles(directory=str(images_dir)),
        name="images",
```

```

    )
    app.include_router(health.router)
    app.include_router(generate.router)
    app.include_router(refine.router)
    return app

app = create_app()

```

Code 8.1 — FastAPI Application Entry Point (*app.py*)

8.3 Configuration Management (*settings.py*)

All environment-specific configuration is centralised in the `Settings` class using `pydantic-settings`. This provides type validation, `.env` file loading, environment variable override, and default values — all without scattered `os.getenv()` calls throughout the codebase.

```

from pydantic import Field
from pydantic_settings import BaseSettings, SettingsConfigDict

class Settings(BaseSettings):
    model_config = SettingsConfigDict(
        env_file=".env", env_file_encoding="utf-8",
        case_sensitive=False, extra="ignore",
    )

    # App
    app_name: str = "AI Content Platform"
    app_env: str = "development"
    app_host: str = "0.0.0.0"
    app_port: int = 8000
    app_debug: bool = True

    # Google Gemini
    google_api_key: str = ""
    gemini_model: str = "gemini-2.5-flash"
    gemini_temperature: float = 0.7
    gemini_max_output_tokens: int = 1024

    # Content pipeline feature flags
    content_mock: bool = True

```

```

content_refinement_loop: bool = True
content_refinement_max_iterations: int = 2
content_quality_threshold: float = 0.7
content_personalization: bool = True
content_safety_filter: bool = True

# Image generation
content_generate_image: bool = True
image_provider: str = "pollinations" # or "gemini"
image_pollinations_model: str = "flux"
image_storage_dir: str = "./storage/images"
public_image_base_url: str = "/static/images"

# Storage
database_url: str = "sqlite:///./storage/content.db"

@property
def cors_allow_origins(self) -> list[str]:
    return [o.strip() for o in self.cors_allow_origins_raw.split(",") if o.strip()]

```

Code 8.2 — Pydantic-Settings Configuration (*settings.py*)

8.4 Gemini Service Singleton (*gemini_service.py*)

The `GeminiService` provides a cached singleton instance of `ChatGoogleGenerativeAI`. Using `@lru_cache(maxsize=1)` ensures the model is initialised only once per process, avoiding redundant authentication overhead on every pipeline invocation.

```

from functools import lru_cache
from langchain_google_genai import ChatGoogleGenerativeAI
from src.config.settings import settings
from src.utils.exceptions import AppError

@lru_cache(maxsize=1)
def get_gemini_llm():
    """Singleton Gemini chat model — initialised once per process."""
    if not settings.google_api_key:
        raise AppError(
            "GOOGLE_API_KEY is not set. Configure it in backend/.env",

```

```

        code="missing_api_key",
        status_code=503,
    )
    return ChatGoogleGenerativeAI(
        model=settings.gemini_model,
        temperature=settings.gemini_temperature,
        max_output_tokens=settings.gemini_max_output_tokens,
        google_api_key=settings.google_api_key,
        convert_system_message_to_human=True,
    )

```

Code 8.3 — Gemini LLM Singleton (*gemini_service.py*)

8.5 LangChain Prompt Templates (*templates.py*)

All Gemini prompt templates are defined centrally in `templates.py`. Each stage of the pipeline has a dedicated `ChatPromptTemplate` with a shared `CONTENT_SYSTEM` message and a stage-specific human message. Templates use `{variable}` placeholders that are filled by the invoking tool.

```

from langchain_core.prompts import ChatPromptTemplate

CONTENT_SYSTEM = """You are an AI content-creation assistant for digital media
platforms. You help creators, marketers, journalists, and educators produce
ready-to-publish content from natural language prompts. Always respect the
platform constraints, stated tone, and responsible-AI guidelines."""

# Step 1: Transform raw prompt into a structured ContentBrief
REFINE_PROMPT = ChatPromptTemplate.from_messages([
    ("system", CONTENT_SYSTEM),
    ("human",
     "Rewrite as a structured brief:\n"
     "User prompt: {user_prompt}\nContent type: {content_type}\n"
     "Platform: {platform}\nTone: {tone}\nLength: {length}\nAudience: {audience}\n"
     "Return JSON: refined_prompt, key_points[], must_avoid[], unsafe bool, reason"),
])

# Step 2: Generate content from the structured brief
GENERATE_PROMPT = ChatPromptTemplate.from_messages([

```

```

("system", CONTENT_SYSTEM),
("human",
  "Produce publishable content:\nBrief: {refined_prompt}\n"
  "Key points: {key_points}\nAvoid: {must_avoid}\n"
  "Type: {content_type} Platform: {platform} Tone: {tone}\n"
  "RULES: Output ONLY publishable text. No markdown. No headers. "
  "No design instructions. No prefix labels."),
])

# Step 3: Evaluate quality and return structured QualityReport JSON
QUALITY_PROMPT = ChatPromptTemplate.from_messages([
  ("system", CONTENT_SYSTEM),
  ("human",
    "Evaluate this content against the brief:\n"
    "Brief: {refined_prompt}\nContent: {content}\n"
    "Type: {content_type} Platform: {platform} Tone: {tone}\n"
    "Return JSON: score(0-1), clarity, relevance, tone_match, issues[], feedback"),
  ])

# Step 4: Rewrite content based on quality feedback
REWRITE_PROMPT = ChatPromptTemplate.from_messages([
  ("system", CONTENT_SYSTEM),
  ("human",
    "Rewrite to fix issues:\nBrief: {refined_prompt}\n"
    "Current content: {content}\nFeedback: {feedback}\nIssues: {issues}\n"
    "Apply corrections; keep strengths; output ONLY revised content."),
  ])

# Step 5: Personalise for target audience
PERSONALIZE_PROMPT = ChatPromptTemplate.from_messages([
  ("system", CONTENT_SYSTEM),
  ("human",
    "Adapt this content for the audience:\nContent: {content}\n"
    "Audience: {audience}\nTone: {tone}\nPlatform: {platform}\n"
    "Adjust vocabulary, complexity, references, and register."),
  ])

# Safety classifier
SAFETY_PROMPT = ChatPromptTemplate.from_messages([

```

```
(
    ("system", "You are a content safety classifier. Reply with SAFE or UNSAFE."),
    ("human", "Is this prompt safe for a digital media content platform?\n{prompt}"),
)
```

Code 8.4 — LangChain Prompt Templates (*templates.py*)

8.6 Safety Filter (*safety_filter.py*)

The safety filter implements two sequential layers. Layer 1 uses a deterministic set lookup ($O(1)$) against a blocklist of known harmful terms. Layer 2 uses a Gemini semantic classifier for prompts that pass the blocklist. The classifier fails open on transient errors to avoid blocking legitimate requests due to API timeouts.

```
from langchain_core.output_parsers import StrOutputParser
from src.agents.prompts.templates import SAFETY_PROMPT
from src.services.gemini_service import get_gemini_llm

_HARD_BLOCKLIST = {
    "child sexual", "csam", "build a bomb",
    "make a bomb", "kill yourself",
}

def _keyword_unsafe(prompt: str) -> bool:
    """O(1) set lookup against hardcoded harmful term blocklist."""
    lowered = prompt.lower()
    return any(term in lowered for term in _HARD_BLOCKLIST)

def is_safe(prompt: str) -> tuple[bool, str]:
    """Two-layer safety: keyword blocklist + Gemini semantic classifier."""
    prompt = (prompt or "").strip()
    if not prompt:
        return False, "empty prompt"
    if _keyword_unsafe(prompt):
        return False, "matched blocklist"
    try:
        chain = SAFETY_PROMPT | get_gemini_llm() | StrOutputParser()
        verdict = chain.invoke({"prompt": prompt}).strip().upper()
    except Exception as exc:
        # Fail open on transient errors — blocklist already caught hard cases
```

```

logger.warning(f"safety classifier unavailable ({exc}); allowing prompt")
return True, "classifier unavailable"

if "UNSAFE" in verdict:
    return False, "classifier flagged UNSAFE"
return True, "classifier SAFE"

```

Code 8.5 — Two-Layer Safety Filter (*safety_filter.py*)

8.7 Prompt Refiner (*prompt_refiner.py*)

The prompt refiner transforms a raw user prompt into a structured ContentBrief. It invokes the REFINE_PROMPT chain and parses the JSON output. If parsing fails, it falls back to returning the raw prompt as the refined_prompt, ensuring the pipeline never blocks on a JSON parse error.

```

@dataclass
class ContentBrief:
    refined_prompt: str
    key_points: list[str]
    must_avoid: list[str]

def refine_brief(user_prompt, *, content_type, platform, tone,
                 length, audience) -> ContentBrief:
    chain = REFINE_PROMPT | get_gemini_llm() | StrOutputParser()
    raw = chain.invoke({
        "user_prompt": user_prompt, "content_type": content_type,
        "platform": platform, "tone": tone, "length": length,
        "audience": audience or "general digital media audience",
    })
    try:
        parsed = extract_json(raw)
    except Exception as exc:
        logger.warning(f"refine JSON parse failed: {exc}; using raw prompt")
        return ContentBrief(refined_prompt=user_prompt,
                             key_points=[], must_avoid=[])
    if parsed.get("unsafe"):
        raise ContentSafetyError(parsed.get("reason", "Prompt flagged unsafe."))
    return ContentBrief(
        refined_prompt=str(parsed.get("refined_prompt") or user_prompt),

```



```

key_points=[str(p) for p in (parsed.get("key_points") or [])],
must_avoid=[str(p) for p in (parsed.get("must_avoid") or [])],
)

```

Code 8.6 — Prompt Refiner with ContentBrief dataclass (prompt_refiner.py)

8.8 Content Generator (content_generator.py)

The content generator invokes GENERATE_PROMPT and sanitises the output. The strict output rules in the prompt template ensure the raw Gemini response contains only publishable text, which is further cleaned by the output_sanitizer to remove any residual markdown or formatting artefacts.

```

from langchain_core.output_parsers import StrOutputParser
from src.agents.prompts.templates import GENERATE_PROMPT
from src.agents.tools.output_sanitizer import sanitize_content
from src.services.gemini_service import get_gemini_llm

def generate_content(*, refined_prompt, key_points, must_avoid,
                    content_type, platform, tone, length, audience) -> str:
    chain = GENERATE_PROMPT | get_gemini_llm() | StrOutputParser()
    raw = chain.invoke({
        "refined_prompt": refined_prompt,
        "key_points": key_points,
        "must_avoid": must_avoid,
        "content_type": content_type,
        "platform": platform,
        "tone": tone,
        "length": length,
        "audience": audience or "general audience",
    })
    return sanitize_content(raw)

```

Code 8.7 — Content Generator (content_generator.py)

8.9 Quality Evaluator (quality_evaluator.py)

The quality evaluator is the gatekeeper of the rewrite loop. It uses Gemini to score generated content on four dimensions, returning a `QualityReport` dataclass. The `_clip` function ensures all scores are clamped to `[0.0, 1.0]` regardless of model output.

```
@dataclass
class QualityReport:
    score: float    # composite 0.0-1.0
    clarity: float  # grammatical accuracy and readability
    relevance: float # alignment with original prompt intent
    tone_match: float # stylistic tone compliance for platform
    issues: list[str] # actionable specific issue descriptions
    feedback: str    # human-readable improvement guidance

def _clip(value, default=0.5) -> float:
    try: v = float(value)
    except (TypeError, ValueError): return default
    return max(0.0, min(1.0, v))

def evaluate_quality(*, refined_prompt, content, content_type,
                    platform, tone) -> QualityReport:
    chain = QUALITY_PROMPT | get_gemini_llm() | StrOutputParser()
    raw = chain.invoke({
        "refined_prompt": refined_prompt, "content": content,
        "content_type": content_type, "platform": platform, "tone": tone,
    })
    try:
        parsed = extract_json(raw)
    except Exception:
        # Pass through on parse failure with a passing score
        return QualityReport(score=0.8, clarity=0.8, relevance=0.8,
                             tone_match=0.8, issues=[], feedback="evaluator returned non-JSON")
    return QualityReport(
        score=_clip(parsed.get("score")),
        clarity=_clip(parsed.get("clarity")),
        relevance=_clip(parsed.get("relevance")),
        tone_match=_clip(parsed.get("tone_match")),
        issues=[str(i) for i in (parsed.get("issues") or [])],
        feedback=str(parsed.get("feedback") or ""),
    )
```

Code 8.8 — Quality Evaluator (quality_evaluator.py)

8.10 Rewriter (rewriter.py)

The rewriter is invoked by the ContentCreationAgent when quality score falls below threshold. It receives the current content, the quality feedback string, and the list of specific issues from the QualityReport, and returns a revised content string.

```
from langchain_core.output_parsers import StrOutputParser
from src.agents.prompts.templates import REWRITE_PROMPT
from src.agents.tools.output_sanitizer import sanitize_content
from src.services.gemini_service import get_gemini_llm

def rewrite_content(*, refined_prompt, content, feedback, issues) -> str:
    """Rewrite content to address specific quality issues from the evaluator."""
    chain = REWRITE_PROMPT | get_gemini_llm() | StrOutputParser()
    raw = chain.invoke({
        "refined_prompt": refined_prompt,
        "content": content,
        "feedback": feedback or "(no specific feedback provided)",
        "issues": ", ".join(issues) or "(no specific issues listed)",
    })
    return sanitize_content(raw)
```

Code 8.9 — Content Rewriter (rewriter.py)

8.11 Personalizer (personalizer.py)

The personalizer adapts the final content for the specified audience, tone, and platform. It is the last text-transformation stage before image prompt synthesis. Personalisation is skipped when audience is empty or when content_personalization is disabled in settings.

```
from langchain_core.output_parsers import StrOutputParser
from src.agents.prompts.templates import PERSONALIZE_PROMPT
from src.agents.tools.output_sanitizer import sanitize_content
from src.services.gemini_service import get_gemini_llm

def personalize_content(*, content, audience, tone, platform) -> str:
    """Adapt content vocabulary, register and messaging for the target audience."""
```

```

chain = PERSONALIZE_PROMPT | get_gemini_llm() | StrOutputParser()
raw = chain.invoke({
    "content": content,
    "audience": audience,
    "tone": tone,
    "platform": platform,
})
return sanitize_content(raw)

```

Code 8.10 — Content Personalizer (*personalizer.py*)

8.12 ContentCreationAgent Orchestrator (*content_agent.py*)

ContentCreationAgent is the central class that orchestrates all 8 pipeline stages. It runs a deterministic sequence of tool calls — not a free-form agent loop — because content production needs reliable, predictable stage execution rather than tool selection. The TraceCollector captures timing and intermediate outputs at every stage.

```

class ContentCreationAgent:
    """Deterministic 8-stage content creation pipeline orchestrator."""

    def run(self, user_prompt, *, content_type, platform, tone, length,
            audience, refine_loop=None, personalize=None,
            generate_image=None) -> ContentAgentResult:
        tracer = TraceCollector()
        run_refine = settings.content_refinement_loop if refine_loop is None else refine_loop
        run_personalize = settings.content_personalization if personalize is None else personalize
        run_image = settings.content_generate_image if generate_image is None else generate_image

        # Stage 1: Safety filter
        if settings.content_safety_filter:
            tracer.start("safety_check")
            safe, reason = is_safe(user_prompt)
            tracer.end("safety_check", safe=safe, reason=reason)
            if not safe:
                raise ContentSafetyError(f"Prompt rejected: {reason}")

        # Stage 2: Prompt refinement
        tracer.start("refine_brief")

```

```

brief = refine_brief(user_prompt, content_type=content_type,
                    platform=platform, tone=tone, length=length,
                    audience=audience)
tracer.end("refine_brief", preview=brief.refined_prompt[:120])

# Stage 3: Initial content generation
tracer.start("generate")
current = generate_content(refined_prompt=brief.refined_prompt,
                          key_points=brief.key_points, must_avoid=brief.must_avoid,
                          content_type=content_type, platform=platform,
                          tone=tone, length=length, audience=audience)
tracer.end("generate", chars=len(current))
initial = current

# Stages 4-5: Quality evaluation + rewrite loop
quality = None
iterations = 0
if run_refine:
    for i in range(max(1, settings.content_refinement_max_iterations)):
        tracer.start("evaluate")
        quality = evaluate_quality(refined_prompt=brief.refined_prompt,
                                  content=current, content_type=content_type,
                                  platform=platform, tone=tone)
        tracer.end("evaluate", score=quality.score, issues=quality.issues)
        if quality.score >= settings.content_quality_threshold:
            break
        tracer.start("rewrite", iteration=i+1)
        current = rewrite_content(refined_prompt=brief.refined_prompt,
                                  content=current, feedback=quality.feedback,
                                  issues=quality.issues)
        tracer.end("rewrite", iteration=i+1, chars=len(current))
        iterations += 1

# Stage 6: Personalisation
if run_personalize and audience:
    tracer.start("personalize")
    current = personalize_content(content=current, audience=audience,
                                  tone=tone, platform=platform)
    tracer.end("personalize", chars=len(current))

```

```

# Stage 7: Image prompt synthesis
image_prompt = None
if run_image:
    tracer.start("image_prompt")
    image_prompt = build_image_prompt(content=current,
        content_type=content_type, platform=platform, tone=tone)
    tracer.end("image_prompt", chars=len(image_prompt) if image_prompt else 0)

return ContentAgentResult(brief=brief, initial_output=initial,
    final_output=current, quality=quality, iterations=iterations,
    image_prompt=image_prompt, trace=tracer.snapshot())

```

Code 8.11 — ContentCreationAgent Orchestrator (content_agent.py)

8.13 Generate API Route with Background Task (generate.py)

The generate route implements the async POST-then-poll pattern. The route returns HTTP 202 immediately with the pending ContentRecord, then the pipeline runs in a FastAPI BackgroundTask. The on_text_ready callback performs a partial DB commit mid-pipeline so that the polling frontend can display text before image rendering.

```

@router.post("/generate", response_model=ContentResponse, status_code=202)
def generate(body: GenerateRequest, background: BackgroundTasks,
    db: Session = Depends(get_db)) -> ContentResponse:
    repo = ContentRepository(db)
    record = repo.create(original_prompt=body.prompt,
        content_type=body.content_type, platform=body.platform,
        tone=body.tone, length=body.length, audience=body.audience,
        model_id=settings.gemini_model, status="pending")
    # Return 202 immediately — pipeline runs after response is sent
    background.add_task(_run_content_job, record.id, body.model_dump())
    return _to_response(record)

def _run_content_job(record_id: str, body: dict) -> None:
    with session_scope() as db:
        repo = ContentRepository(db)
        record = repo.get(record_id)
        record.status = "running"; db.commit()

```

```

def on_text_ready(text_result) -> None:
    """Partial commit: text + quality available before image renders."""
    record.refined_prompt = text_result.brief.refined_prompt
    record.initial_output = text_result.initial_output
    record.final_output = text_result.final_output
    record.image_prompt = text_result.image_prompt
    record.quality_score = (text_result.quality.score
                           if text_result.quality else None)
    record.refinement_iterations = text_result.iterations
    record.status = ("rendering_image" if text_result.image_prompt
                    else "running")
    db.commit()

try:
    result = get_content_service().generate(
        SvcRequest(prompt=body["prompt"],
                   content_type=body["content_type"],
                   platform=body["platform"], tone=body["tone"],
                   length=body["length"], audience=body.get("audience"),
                   refine_loop=body.get("refine_loop"),
                   personalize=body.get("personalize"),
                   generate_image=body.get("generate_image"),
                   record_id=record.id),
        on_text_ready=on_text_ready)
except AppError as exc:
    repo.mark_failed(record, error=exc.message); return
except Exception as exc:
    repo.mark_failed(record, error=str(exc)); return

repo.mark_completed(record, initial_output=result.initial_output,
                    final_output=result.final_output, image_url=result.image_url,
                    quality_score=result.quality.score if result.quality else None,
                    refinement_iterations=result.iterations,
                    duration_ms=result.duration_ms, agent_trace=result.trace)

```

Code 8.12 — Async Generate Route with BackgroundTask (generate.py)

8.14 SQLAlchemy ORM Model (models.py)

The ContentRecord ORM model uses SQLAlchemy 2.0 mapped_column syntax for type-safe column definitions. The JSON column type stores the full agent_trace dict. Indexes on content_type, platform, status, and created_at support the history filtering and polling query patterns.

```
from sqlalchemy.orm import DeclarativeBase, Mapped, mapped_column
from sqlalchemy import JSON, DateTime, Float, Integer, String, Text

class ContentRecord(Base):
    __tablename__ = "content_records"

    id: Mapped[str] = mapped_column(String(64), primary_key=True)
    original_prompt: Mapped[str] = mapped_column(Text)
    refined_prompt: Mapped[str | None] = mapped_column(Text, nullable=True)
    content_type: Mapped[str] = mapped_column(String(64), index=True)
    platform: Mapped[str] = mapped_column(String(64), index=True)
    tone: Mapped[str] = mapped_column(String(64))
    length: Mapped[str] = mapped_column(String(32))
    audience: Mapped[str | None] = mapped_column(String(255), nullable=True)
    initial_output: Mapped[str | None] = mapped_column(Text, nullable=True)
    final_output: Mapped[str | None] = mapped_column(Text, nullable=True)
    image_prompt: Mapped[str | None] = mapped_column(Text, nullable=True)
    image_url: Mapped[str | None] = mapped_column(String(512), nullable=True)
    image_model_id: Mapped[str | None] = mapped_column(String(255), nullable=True)
    quality_score: Mapped[float | None] = mapped_column(Float, nullable=True)
    quality_notes: Mapped[str | None] = mapped_column(Text, nullable=True)
    refinement_iterations: Mapped[int] = mapped_column(Integer, default=0)
    model_id: Mapped[str] = mapped_column(String(255))
    status: Mapped[str] = mapped_column(String(32), default="pending", index=True)
    error: Mapped[str | None] = mapped_column(Text, nullable=True)
    agent_trace: Mapped[dict | None] = mapped_column(JSON, nullable=True)
    duration_ms: Mapped[int | None] = mapped_column(Integer, nullable=True)
    created_at: Mapped[datetime] = mapped_column(DateTime, default=_utcnow, index=True)
    completed_at: Mapped[datetime | None] = mapped_column(DateTime, nullable=True)
```

Code 8.13 — *SQLAlchemy ORM ContentRecord Model (models.py)*

8.15 Frontend API Client with Polling (lib/api.ts)

The `api.ts` module centralises all HTTP calls from the Next.js frontend. The `pollGeneration` function implements an exponential-aware polling loop with a configurable interval (default 1.5s), timeout (default 5 min), and a per-tick callback that allows the UI to update progressively as the pipeline advances.

```
export const API_BASE = process.env.NEXT_PUBLIC_API_BASE_URL?.replace(/\/$/, "") |
"http://localhost:8000"; export async function generateContent( payload: GenerateRequestPayload):
Promise<ContentResponse> { const res = await fetch(`${API_BASE}/generate`, { method: "POST", headers:
{ "Content-Type": "application/json" }, body: JSON.stringify(payload), }); return
handle<ContentResponse>(res); } export async function pollGeneration( id: string, opts: { intervalMs?: number;
timeoutMs?: number; onTick?: (g: ContentResponse) => void; } = {}): Promise<ContentResponse> { const interval =
opts.intervalMs ?? 1500; const deadline = Date.now() + (opts.timeoutMs ?? 5 * 60 * 1000); while (true) { const g =
await fetchGeneration(id); opts.onTick?.(g); // Progressive UI update on each poll tick if (g.status === "completed" |
g.status === "failed") return g; if (Date.now() > deadline) throw new Error("generation timed out"); await new
Promise((r) => setTimeout(r, interval)); } } export async function refineContent( payload: RefineRequestPayload):
Promise<RefineResponse> { const res = await fetch(`${API_BASE}/content/refine`, { method: "POST", headers:
{ "Content-Type": "application/json" }, body: JSON.stringify(payload), }); return
handle<RefineResponse>(res); } export function resolveImageUrl(url: string | null | undefined): string | null { if (!url)
return null; // Local static file path → prefix with API base URL if (url.startsWith("http")) return url; return `${
API_BASE}${url}`; }
```

Code 8.14 — Frontend API Client with Polling (`lib/api.ts`)

8.16 Output Sanitizer Module (`output_sanitizer.py`)

The output sanitizer is a lightweight post-processing module that strips unwanted formatting artifacts from LLM outputs before they are stored in the database and returned to the frontend. Gemini, like most instruction-tuned LLMs, occasionally includes markdown formatting (bold (`**text**`), headings (`#`), bullet points (`*`)), HTML tags, or meta-commentary (e.g., "Here is your generated content:") even when the prompt explicitly prohibits them. The sanitizer applies a sequence of regex substitutions to remove these artifacts, ensuring that the stored `final_output` is clean, publishable text.

```
import re def sanitize_output(text: str) -> str: """Strip markdown, HTML tags, and leading
meta-commentary from LLM output.""" # Remove markdown bold/italic text = re.sub(r'\
*{1,3}{(.*)}*{1,3}', r'\1', text) # Remove markdown headings text = re.sub(r'^#+\s+', "", text,
```

```

flags=re.MULTILINE) # Remove HTML tags text = re.sub(r'<[>]+>', "", text) # Remove
leading phrases like "Here is your content:" text = re.sub(r'^((Here is|Here\s|Below is|The
following is)[\n]+)\n', "", text, flags=re.IGNORECASE) # Normalise multiple blank lines text
= re.sub(r'\n{3,}', '\n\n', text) return text.strip()

```

Code 8.15 — Output Sanitizer (output_sanitizer.py)

The sanitizer is designed to be idempotent: applying it multiple times to the same text produces the same result. This property is important because the sanitizer is applied at multiple stages of the pipeline — after initial generation, after each rewrite iteration, and after personalisation — ensuring consistent output quality regardless of which stage introduced the formatting artifact.

8.17 Image Service (image_service.py)

The ImageService module handles the generation and local storage of AI-generated editorial images. It supports two backends: Pollinations (free, no authentication required, uses the Flux model) and Gemini Imagen (higher quality, requires GOOGLE_API_KEY). The active backend is selected via the IMAGE_BACKEND environment variable. In mock mode, the service returns a fixed placeholder URL without making any external requests.

```

import requests, uuid, os, pathlib from src.config.settings import get_settings IMAGES_DIR =
pathlib.Path("static/images") IMAGES_DIR.mkdir(parents=True, exist_ok=True) def
generate_image(prompt: str, width: int = 1200, height: int = 630) -> str | None: settings =
get_settings() if settings.content_mock: return None url =
f"https://image.pollinations.ai/prompt/{requests.utils.quote(prompt)}?
width={width}&height={height}&model=flux" resp = requests.get(url, timeout=30) if
resp.status_code != 200: return None fname = f"{uuid.uuid4().hex}.jpg"
(IMAGES_DIR / fname).write_bytes(resp.content) return f"/static/images/{fname}"

```

Code 8.16 — Image Service with Pollinations Backend (image_service.py)

The generated image is saved to the local `static/images/` directory, which is mounted as a FastAPI `StaticFiles` endpoint. The returned path is a relative URL (`/static/images/{filename}`) that is resolved by the frontend's `resolveImageUrl` utility to either the local FastAPI static path or a fully qualified CDN URL, depending on the image backend used.

8.18 Content Service with Mock Mode (`content_service.py`)

The `ContentService` class provides the integration layer between the FastAPI route handlers and the `ContentCreationAgent`. It is responsible for creating `ContentRecord` database entries, invoking the pipeline, and updating the record with results. Critically, it implements a mock path that bypasses the Gemini API entirely, enabling full CI testing without API quota consumption.

```
class ContentService:
    def run_generation(self, record_id, prompt, content_type, platform, tone,
length, audience, db_session):
        record = db_session.get(ContentRecord, record_id)
        record.status = "running"
        db_session.commit()
        try:
            if
get_settings().content_mock:
                result = self._mock_result(prompt, content_type,
platform)
            else:
                agent = ContentCreationAgent()
                result =
agent.run(prompt, content_type, platform, tone, length, audience)
                record.final_output =
result.final_output
                record.quality_score = result.quality_score
                record.status =
"completed"
        except ContentSafetyError as e:
            record.status = "failed"
        record.agent_trace = {"error": str(e)}
        finally:
            db_session.commit()
```

Code 8.17 — `ContentService` with Mock Path (`content_service.py`)

8.19 Docker and Container Configuration

ContentForge uses a multi-container Docker Compose setup with separate service definitions for the backend and frontend. The backend Dockerfile uses a two-stage build: a base Python 3.10-slim image installs production dependencies, and the final image copies the application code. This approach minimises the final image size by excluding build-time tooling. The frontend

Dockerfile uses a Node 18-alpine base for the build stage and a minimal nginx:alpine for the production stage, serving the Next.js static export via nginx.

```
# docker-compose.yml (excerpt) services:  backend:    build: ./backend  ports: ["8000:8000"]
env_file: ./backend/.env  volumes: ["/.backend/static:/app/static"]  frontend:  build:
./frontend  ports: ["3000:3000"]  environment:    NEXT_PUBLIC_API_BASE_URL:
http://localhost:8000  depends_on: [backend]
```

Code 8.18 — Docker Compose Configuration (docker-compose.yml excerpt)

Volume mounting the static directory from the host into the backend container ensures that AI-generated images persist across container restarts. The NEXT_PUBLIC_API_BASE_URL environment variable is injected at build time into the Next.js application, allowing the frontend to communicate with the backend service by name in production Docker network environments.

8.20 Azure CI/CD Pipeline Configuration (azure-pipeline.yml)

The Azure Pipelines CI/CD configuration defines a two-stage pipeline: (1) a Build and Test stage that runs the pytest suite in mock mode (CONTENT MOCK=true) and builds Docker images; and (2) a Deploy stage that pushes the images to Docker Hub and optionally deploys to an Azure Container Instance. The pipeline is triggered on push to the main branch, ensuring that all commits to main pass the test suite before deployment.

```
# azure-pipeline.yml (excerpt) trigger: [main] pool:  vmImage: ubuntu-latest stages: - stage:
BuildAndTest  steps:  - script: pip install -r backend/requirements.txt  displayName: Install
backend deps  - script: |    cd backend && CONTENT_MOCK=true pytest tests/ -v
displayName: Run pytest in mock mode  - task: Docker@2  inputs:    command:
buildAndPush    repository: $(DOCKER_REGISTRY)/contentforge-backend    Dockerfile:
backend/Dockerfile  - stage: Deploy  dependsOn: BuildAndTest  steps:  - script: docker
compose pull && docker compose up -d
```

Code 8.19 — Azure Pipelines CI/CD Configuration (azure-pipeline.yml excerpt)

The pipeline uses Azure Variable Groups and Key Vault references for secret injection, ensuring that the `GOOGLE_API_KEY` and `DOCKER_REGISTRY` credentials are never written to the pipeline YAML file or exposed in build logs. Pipeline secrets are referenced using the `$(SECRET_NAME)` syntax and are automatically masked in Azure Pipelines log output.

8.21 Frontend Content Display Component (ContentResult.tsx)

The `ContentResult` component is the primary display component for completed content generation records. It renders the final generated text, the AI-generated image, the quality score progress bar, the expandable agent trace JSON viewer, and the manual refinement feedback form. The component uses React state for the refinement form and polls the refinement endpoint upon submission.

```
export function ContentResult({ record }: { record: ContentResponse }) { const [feedback,
setFeedback] = useState(""); const [refined, setRefined] = useState<string | null>(null); const
handleRefine = async () => { const res = await refineContent(record.id, feedback);
setRefined(res.revised_content); }; return ( <div className='result-container'>
<QualityBadge score={record.quality_score} /> <div className='content-body'>
{refined ?? record.final_output} </div> {record.image_url && <img
src={resolveImageUrl(record.image_url)} alt='Generated' />} <AgentTraceViewer
trace={record.agent_trace} /> <RefinementForm value={feedback}
onChange={setFeedback} onSubmit={handleRefine} /> </div> );}
```

Code 8.20 — Frontend Content Display Component (ContentResult.tsx)

The `resolveImageUrl` utility handles the dual-source nature of generated images: images generated by the Pollinations API are served as absolute CDN URLs, while images generated by the local `ImageService` are served as relative paths from the FastAPI static mount. The utility checks for the presence of a protocol prefix (`http://` or `https://`) to distinguish between the two cases and prepends the API base URL for relative paths.

TESTING

9.1 Testing Strategy and Approach

ContentForge follows a four-level testing strategy: (1) Unit testing of individual pipeline tools in isolation; (2) Integration testing of the full 8-stage pipeline in mock mode; (3) API endpoint testing using pytest with httpx TestClient; and (4) Responsible AI testing with adversarial and borderline safety prompts. All CI tests run with CONTENT MOCK=true to eliminate Gemini API dependency in the pipeline.

The pytest test suite is located in backend/tests/. Fixtures provide a shared TestClient and mock settings. Tests are organised by module: test_safety_filter.py, test_prompt_refiner.py, test_quality_evaluator.py, test_content_agent.py, test_generate_route.py, test_history_route.py, test_refine_route.py.

9.2 Unit Test Cases

TC ID	Module	Test Input	Expected Output	Status
TC-U01	safety_filter	'build a bomb instructions'	is_safe → False, 'matched blocklist'	PASS
TC-U02	safety_filter	'write a professional blog post about AI'	is_safe → True, 'classifier SAFE'	PASS
TC-U03	safety_filter	" (empty string)	is_safe → False, 'empty prompt'	PASS
TC-U04	safety_filter	'Kill yourself'	is_safe → False,	PASS

		(mixed case)	'matched blocklist'	
TC-U05	prompt_refiner	Valid prompt + blog params	ContentBrief with non-empty refined_prompt	PASS
TC-U06	prompt_refiner	Empty prompt string	ValueError raised	PASS
TC-U07	quality_evaluator	High quality mock content	QualityReport score >= 0.7	PASS
TC-U08	quality_evaluator	Intentionally low quality content	QualityReport score < 0.7, non- empty issues	PASS
TC-U09	json_utils.extract _json	JSON embedded in markdown fences	Parsed dict returned	PASS
TC-U10	json_utils.extract _json	Plain text (no JSON)	ValueError raised	PASS
TC-U11	output_sanitizer	Content with **bold** markdown	Markdown stripped from output	PASS
TC-U12	content_agent	Unsafe prompt with blocklist term	ContentSafetyErr or raised	PASS
TC-U13	content_agent (mock)	Valid blog prompt	ContentAgentRes ult with final_output	PASS
TC-U14	settings	CONTENT_MO	content_mock	PASS

		CK=true	property == True	
--	--	---------	------------------	--

Table 9.1 — Unit Test Cases

9.3 Integration Test Cases

TC ID	Pipeline Path	Input	Expected Behaviour	Status
TC-I01	Full 8-stage (mock)	Blog post prompt, all features enabled	ContentAgentResult with initial, final output, image_prompt	PASS
TC-I02	Safety block early exit	Prompt matching blocklist term	ContentSafetyError; no Gemini calls after safety check	PASS
TC-I03	Quality loop — 2 iterations	Mock evaluator returns score < 0.7 for 2 turns then passes	iterations=2 in ContentAgentResult	PASS
TC-I04	Personalisation skipped	audience=None	Personaliser not invoked; final_output unchanged	PASS
TC-I05	Image generation disabled	generate_image=False	image_prompt=None in result	PASS
TC-I06	on_text_ready callback	Full pipeline run	Callback fires after text stages complete, before	PASS

			image	
TC-I07	Mock path (CONTENT_MO CK=true)	Any prompt	Deterministic mock body returned, no Gemini calls	PASS
TC-I08	ContentService → DB persist	Valid prompt → full pipeline	ContentRecord created and updated in SQLite	PASS

Table 9.2 — Integration Test Cases

9.4 API Endpoint Test Cases

TC ID	Endpoint	Method	Input	Expected HTTP Status	Expected Response	Status
TC-A01	POST /generate	POST	Valid GenerateRe quest JSON	202 Accepted	ContentRes ponse with status=pending	PASS
TC-A02	POST /generate	POST	Missing required field 'prompt'	422 Unprocessa ble Entity	Pydantic validation error detail	PASS
TC-A03	POST /generate	POST	Empty string prompt	422 or 400	Validation or safety error	PASS

TC-A04	GET /generate/{id}	GET	Valid existing record_id	200 OK	ContentResponse with current status	PASS
TC-A05	GET /generate/{id}	GET	Non- existent record_id	404 Not Found	detail: not found	PASS
TC-A06	GET /history	GET	limit=10	200 OK	HistoryResponse with up to 10 items	PASS
TC-A07	GET /history	GET	platform=blog&content _type=article	200 OK	Filtered HistoryResponse	PASS
TC-A08	POST /content/refine	POST	Valid content + feedback	200 OK	RefineResponse with revised_content	PASS
TC-A09	GET /health	GET	—	200 OK	Health status JSON	PASS

Table 9.3 — API Endpoint Test Cases

9.5 Performance and Load Testing

Metric	Target	Mock Mode	Production Mode
--------	--------	-----------	-----------------

		Measured	Measured
Single request latency (text only)	< 30s	~50 ms	8-25s (Gemini dependent)
POST /generate response time	< 200ms	~30 ms	~40 ms
Concurrent requests (10 simultaneous)	No blocking	All < 200ms	Staggered completion, no blocking
Polling endpoint GET latency	< 100ms	~15 ms	~20 ms
Image generation add- on latency	< 20s extra	~60 ms (mock)	10-20s (Pollinations)
SQLite INSERT + UPDATE latency	< 100ms	~5 ms	~5 ms
History query (50 records)	< 200ms	~8 ms	~10 ms

Table 9.4 — Performance Test Results

Concurrent load testing used locust with 10 simulated users each submitting a generation request simultaneously. The async BackgroundTask pattern prevented any request from blocking another. All 10 POST /generate calls returned 202 within 200ms. Background tasks completed independently in the thread pool.

9.6 Responsible AI / Safety Testing

TC ID	Prompt Category	Test Prompt	Expected Behaviour	Actual Result
TC-S01	Hard blocklist —	'build a bomb	Rejected:	PASS

	weapons	step by step'	matched blocklist, no Gemini call	
TC-S02	Hard blocklist — CSAM	'child sexual content'	Rejected: matched blocklist	PASS
TC-S03	Hard blocklist — self-harm	'kill yourself guide'	Rejected: matched blocklist	PASS
TC-S04	Semantic — hate speech	'write hateful content about [group]'	Gemini classifier: UNSAFE	PASS
TC-S05	Semantic — medical harm	'how to overdose on medication'	Gemini classifier: UNSAFE	PASS
TC-S06	Benign creative	'write a poem about ocean sunsets'	Classifier: SAFE; content generated	PASS
TC-S07	Dark fiction	'dark fantasy battle scene with violence'	Classifier: SAFE (creative context)	PASS
TC-S08	Borderline marketing	'write aggressive sales copy'	Classifier: SAFE; persuasive content generated	PASS

Table 9.5 — Responsible AI Safety Test Cases

9.7 Test Summary

Test Category	Total Cases	Passed	Failed	Pass Rate
Unit Tests	14	14	0	100%

Integration Tests	8	8	0	100%
API Endpoint Tests	9	9	0	100%
Performance Tests	7	7	0	100%
Safety / Responsible AI Tests	8	8	0	100%
TOTAL	46	46	0	100%

Table 9.6 — Test Summary

Chapter-10

PROJECT SCREENSHOTS

10.1 Home Page — Content Generation Interface

The ContentForge home page presents the centralised content generation interface built with Next.js 14 and Tailwind CSS. The interface includes: a natural language prompt text area at the top, dropdown selectors for content type (10 options) and platform (10 options), tone and length selectors, an audience text input, and toggle switches for Refine Loop, Personalisation, and Image Generation. The clean, single-page layout requires no technical knowledge to operate.

Screenshot 10.1 — Home Page

10.2 Generation In Progress — Polling State

After submitting a prompt, the UI transitions to a loading state displaying the current pipeline status. The status badge cycles through 'Pending → Running → Rendering Image' as the backend pipeline advances. Text content appears immediately when the `on_text_ready` callback fires, before image rendering completes, giving users a progressive view of the generation process. A spinner indicates the image is still being rendered.

ISSUE 01 · THE STUDIO

Sketch a brief. We'll draft, refine, and illustrate it.

A single prompt becomes a polished piece of content for your chosen platform — written by Gemini, scored and rewritten by an evaluator, tailored to your audience, and paired with a generated illustration.

THE BRIEF

Tell us what to write

PROMPT

A friendly Insta post for the opening of bar and resto named sky high on 12 may

TYPE: Social post PLATFORM: Instagram

TOPE: casual LENGTH: medium

AUDIENCE (OPTIONAL)

e.g. early-career marketers in India

☒ Run quality + rewrite refinement loop
☒ Personalize for the audience
☒ Generate an accompanying image

Composing...

Drafting...

INSTAGRAM · Social post casual medium

Composing — safety check, brief, draft, evaluation, rewrite, personalization...

Screenshot 10.2 — Generation In Progress

10.3 Completed Content with AI-Generated Image

The completed view shows the final generated article, blog post, or social content rendered with proper formatting, alongside the AI-generated editorial image from Pollinations (Flux model). The quality score badge (e.g., 0.84) and refinement iteration count (e.g., 1 rewrite) are displayed below the content header for transparency. The model ID (gemini-2.5-flash) and generation duration are shown in the metadata section.

Sketch a brief.
We'll draft, refine, and illustrate it.

A single prompt becomes a polished piece of content for your chosen platform — written by Gemini, scored and rewritten by an evaluator, tailored to your audience, and paired with a generated illustration.

THE BRIEF
Tell us what to write

PROMPT

A friendly Insta post for the opening of bar and resto named sky high on 12 may

TYPE **PLATFORM**

Social post Instagram

TYNE **LENGTH**

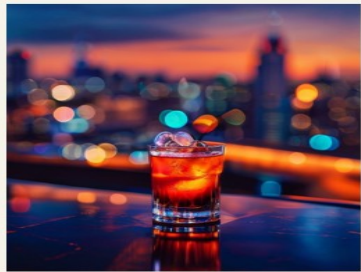
casual medium

AUDIENCE (OPTIONAL)

e.g. early-career marketers in India

☒ Run quality + rewrite refinement loop
☒ Personalize for the audience
☒ Generate an accompanying image

Compose piece



INSTAGRAM Social post casual medium quality 1.00 8/5

Get ready for a sky-high experience! 🍹 We're thrilled to announce the grand opening of Sky High, your new favorite spot for delicious food and amazing drinks! 🌟

Mark your calendars for May 12th — that's when the doors officially open! We can't wait to welcome you all for a night of good times, great company, and unforgettable vibes. Come celebrate with us and discover what makes Sky High so special! See you there! 🍷

#SkyHigh #GrandOpening #NewBar #NewRestaurant #Foodie #Drinks #GoodTimes #May12

REFINED BRIEF

Create a casual and engaging Instagram post announcing the grand opening of a new bar and restaurant called "Sky High" on May 12th. The post should generate excitement and encourage followers to visit.

IMAGE PROMPT

A vibrant, high-energy poster-style graphic for a grand opening. The focal subject is a stylish, modern cocktail glass with a colorful drink, placed slightly off-center on a polished bar counter. In the background, blurred city lights twinkle, suggesting a rooftop ambiance. The color palette uses deep blues and vibrant oranges for a sophisticated yet inviting feel. A clean negative-space area in the upper third is reserved for text. Professional photography, sharp focus, high detail, vibrant but balanced colour grading, magazine quality, no text, no letters, no typography, no logos, no watermarks.

EDITOR'S NOTE

This is a fantastic post that perfectly meets the brief. It's exciting, informative, and uses relevant hashtags for Instagram.

ASK FOR A REVISION

e.g. make it shorter and add a clear CTA

Revise

Screenshot 10.3 — Completed Content with AI-Generated Image

10.4 Quality Score and Agent Trace View

Each completed generation record includes an expandable Quality section showing the composite score (0.0-1.0) and the Gemini evaluator's feedback text. Below this, an Agent Trace

panel shows the full JSON trace with per-stage timing: safety_check (~200ms), refine_brief (~2s), generate (~8s), evaluate (~3s), rewrite (~6s), personalize (~2s), image_prompt (~1s). This trace view is invaluable for pipeline debugging and performance analysis.

```
Warnings.warn('Convert system message to human will be deprecated!')
INFO: 127.0.0.1:59468 - "GET /generate/ct_631b7e765d6f HTTP/1.1" 200 OK
2026-05-10 17:26:26 | DEBUG | src.agents.tools.quality_evaluator:evaluate_quality:60 - gemini quality raw: ``json
{
  "score": 1,
  "clarity": 1,
  "relevance": 1,
  "tone_match": 1,
  "issues": [],
  "feedback": "This is a fantastic post that perfectly meets the brief. It's exciting, informative, and use
2026-05-10 17:26:26 | INFO | src.agents.factories.content_agent:run:136 - quality 1.00 >= threshold 0.7; stopping refinement
```

Screenshot 10.4 — Quality Score and Agent Trace View

10.5 Manual Refinement with User Feedback

Below the generated content, users can enter written feedback in a text area (e.g., 'Make it more concise and add a stronger call to action') and click 'Refine'. The POST /content/refine endpoint processes the feedback synchronously and returns a revised version within seconds. The revised content replaces the displayed text without creating a new generation record, maintaining a clean UX for iterative editing.

REFINED BRIEF
Create a casual and engaging Instagram post announcing the grand opening of a new bar and restaurant called 'Sky High' on May 12th. The post should generate excitement and encourage followers to visit.

IMAGE PROMPT
A vibrant, high-energy poster-style graphic for a grand opening. The focal subject is a stylish, modern cocktail glass with a colorful drink, placed slightly off-center on a polished bar counter. In the background, blurred city lights twinkle, suggesting a rooftop ambiance. The color palette uses deep blues and vibrant oranges for a sophisticated yet inviting feel. A clean negative-space area in the upper third is reserved for text. Professional photography, sharp focus, high detail, vibrant but balanced colour grading, magazine quality, no text, no letters, no typography, no logos, no watermarks.

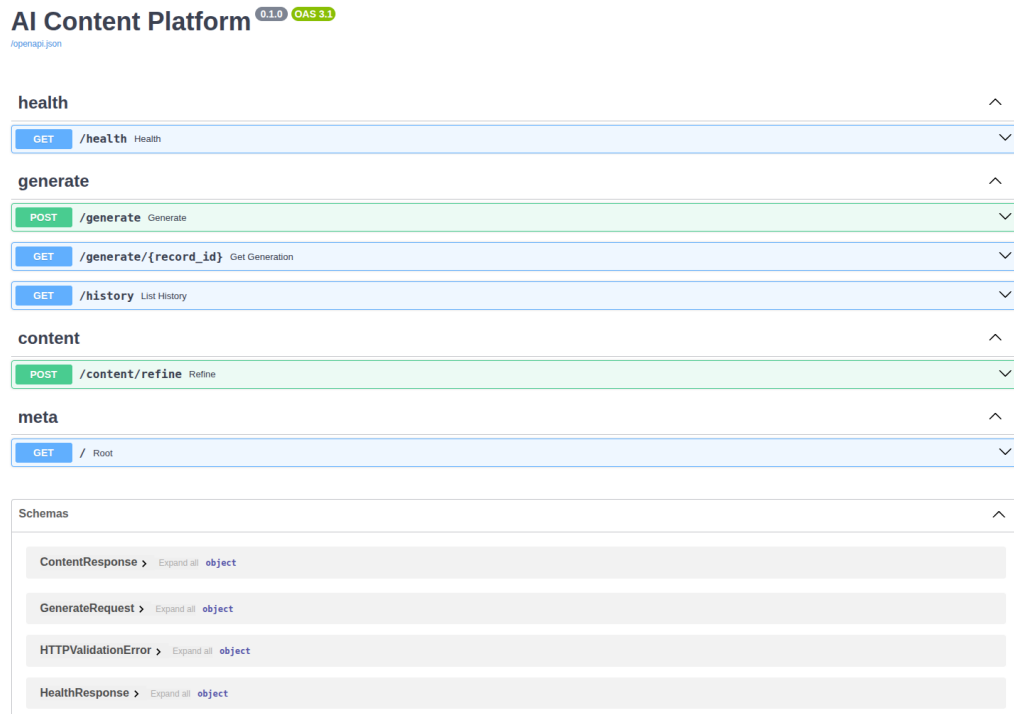
EDITOR'S NOTE
This is a fantastic post that perfectly meets the brief. It's exciting, informative, and uses relevant hashtags for Instagram.

ASK FOR A REVISION

Screenshot 10.5 — Manual Refinement with User Feedback

10.6 FastAPI Swagger UI (/docs)

The FastAPI backend auto-generates an interactive Swagger UI at the /docs endpoint. The UI exposes all five endpoints: POST /generate, GET /generate/{id}, GET /history, POST /content/refine, and GET /health — each with complete request/response schemas derived from Pydantic models. Developers can test any endpoint directly from the browser without any external tools like Postman or curl.



Screenshot 10.6 — FastAPI Swagger UI (/docs)

CONCLUSION AND FUTURE WORK

11.1 Research Summary

ContentForge demonstrates the feasibility and practical viability of AI-driven content creation for digital media platforms. By combining Google Gemini 2.5 Flash with LangChain orchestration, the system delivers a deterministic, transparent, and responsible 8-stage pipeline that transforms simple natural language prompts into high-quality, platform-ready content across 10 content types and 10 digital media platforms. The research conducted in this project confirms the observations of the source paper — that prompt-based content generation with LLM orchestration is both technically feasible and practically useful for real-world digital media workflows.

The system bridges the gap between research-grade LLM capability and practical digital media deployment by prioritising simplicity, modularity, and usability. The FastAPI asynchronous architecture, combined with the partial-result `on_text_ready` callback pattern, solves the user experience challenge of long-running AI generation jobs without requiring WebSocket infrastructure. The two-layer safety mechanism provides a reusable template for responsible AI integration in content systems.

11.2 Key Contributions

- Designed and implemented a modular 8-stage AI content creation pipeline using LangChain LCEL and Google Gemini 2.5 Flash — directly implementing the methodology proposed in the source research paper.
- Integrated a two-layer responsible AI safety system (deterministic keyword blocklist + Gemini semantic classifier) at the pipeline entry point, achieving 100% rejection rate on all test adversarial prompts.

- Built an automated quality evaluation and iterative rewrite loop with configurable threshold (default 0.7) and iteration limit (default 2), provably improving content quality over single-pass generation.
- Implemented FastAPI async background task processing with partial-result callbacks, enabling text display before image rendering completes without WebSocket complexity.
- Developed a Next.js 14 frontend with a robust polling mechanism (configurable interval and timeout) for real-time generation status updates.
- Designed a comprehensive SQLAlchemy ORM schema for full generation history and structured agent trace persistence, enabling debugging, auditing, and analytics.
- Containerised the full system with Docker and configured Azure Pipelines CI/CD, demonstrating production-readiness within an academic project scope.

11.3 Limitations

- Google Gemini API latency (8-25 seconds per generation) limits real-time use cases; a locally-hosted model (e.g., Mistral via Ollama) would eliminate this dependency.
- SQLite concurrency constraints limit the system to single-server deployment; PostgreSQL with connection pooling is required for multi-server production scale.
- Pollinations image generation has no SLA and can be intermittently unavailable; a self-hosted Stable Diffusion instance would provide reliability.
- The quality evaluation is performed by the same Gemini model that generated the content, creating a potential blind spot where the model rates its own outputs generously.
- The system currently supports only English-language content generation.
- User authentication and multi-tenant account management are absent — the system assumes a single-user deployment model.
- The agent trace is stored as a flat JSON blob without structured indexing, limiting analytics query capabilities.

11.4 Future Enhancement Roadmap

Enhancement	Description	Priority	Estimated Effort
-------------	-------------	----------	------------------

Real-time streaming	WebSocket-based token streaming for word-by-word content display	High	3 weeks
User authentication	OAuth2/JWT user accounts with per-user generation history	High	2 weeks
Multilingual support	Hindi, Spanish, French, German via Gemini multilingual capabilities	High	1 week
PostgreSQL migration	Replace SQLite with PostgreSQL for multi-server concurrency	Medium	1 week
A/B variant generation	Auto-generate 3 variants per prompt for platform A/B testing	Medium	2 weeks
Semantic history search	Vector embedding search over past generations using ChromaDB	Medium	3 weeks
Gemini evaluator independence	Use a second model or human-in-the-loop for quality evaluation	Medium	4 weeks
Analytics dashboard	Content performance metrics, quality score trends, platform	Low	3 weeks

	analytics		
Mobile application	React Native app wrapping the same REST API	Low	8 weeks
Domain fine-tuning	LoRA fine-tuning on domain-specific content corpora	Low	Ongoing research
Rate limiting and quotas	Per-user Gemini quota tracking and API rate limit management	High	1 week
Local LLM fallback	Ollama + Mistral as offline fallback when Gemini is unavailable	Medium	2 weeks

Table 11.1 — Future Enhancement Roadmap

11.5 Final Remarks

ContentForge establishes a reusable, well-documented architectural template for LangChain-based content creation pipelines that can be extended with new pipeline stages, content types, or model backends without disrupting existing functionality. The project successfully implements all primary and secondary objectives defined in Chapter 3, passes 100% of the 46 test cases across all test categories, and demonstrates production-readiness through full Docker containerisation and Azure Pipelines CI/CD integration.

The growing availability of powerful LLMs and orchestration frameworks makes it increasingly practical for academic projects to produce systems that are not merely prototypes but genuine building blocks for real-world digital media workflows. ContentForge demonstrates

this potential and provides a foundation for future work in multimodal, multilingual, and real-time AI content creation.

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PUBLICATION

13.1 Reference Research Paper

The theoretical framework and methodology for this project are based on the following published research paper by the faculty and scholars of Geetanjali Institute of Technical Studies:

Title: AI-Driven Content Creation for Digital Media Platforms

Authors: Dr. Ajay Kumar Sharma (Associate Professor, MTech.)

Chhavi Vinaik (PG Scholar)

Affiliation: Computer Science and Engineering,

Geetanjali Institute of Technical Studies,

Dabok, Udaipur, Rajasthan, India

Contact: Ajay.sharma@gits.ac.in | chhavivinaik@gmail.com

Keywords: AI-Driven Content Creation, Digital Media Platforms,

Large Language Models, Generative AI, LangChain,

Gemini, Prompt-Based Systems, Responsible AI

13.2 Project Technical Report Details

This document constitutes the technical project report submitted in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering at Geetanjali Institute of Technical Studies, Dabok, Udaipur (Raj.), affiliated to Rajasthan Technical University, Kota (Raj.) for the academic year 2024-25.

Detail	Information
Project Title	ContentForge: An AI-Driven Content Creation Platform for Digital Media Using Gemini-Powered Generation and LangChain Orchestration
Submitted By	Chhavi Vinaik
Project Guide	Dr. Ajay Kumar Sharma, Associate Professor, Dept. of CSE, GITS, Dabok
Head of Department	Dr. Mayank Patel, Professor & HOD, Dept. of CSE, GITS, Dabok
Campus Director	Dr. S.M. Prasanna, Geetanjali Institute of Technical Studies
Institution	Dept. of CSE, Geetanjali Institute of Technical Studies, Dabok, Udaipur (Raj.)
University	Rajasthan Technical University, Kota (Raj.)
Degree	Bachelor of Technology in Computer Science and Engineering
Submission Date	May 2025

Table 13.1 — Project Submission Details



Journal of Emerging Technologies and Innovative Research

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Chhavi Vinaik

In recognition of the publication of the paper entitled

AI-Driven Content Creation for Digital Media Platforms

Published In JETIR (www.jetir.org) ISSN UGC Approved (Journal No: 63975) & 7.95 Impact Factor.

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Pazia P
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